



UTM
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Faculty of Computing

Network Operations Virtualization for Faculty of Computing Block N28B by NetworkNinjaz

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Table of Content

Introduction.....	6
Project Background.....	7
Task 1.....	9
Floor Plan: Floor 1.....	9
Floor Plan: Floor 2.....	10
Plan Details: General Purpose Lab.....	11
Plan Details: Security Lab.....	12
Plan Details: IoT Lab.....	13
Plan Details: Network Lab.....	14
Plan Details: Video Conference Room.....	15
Reflection.....	16
Task 2.....	17
Topology Design.....	17
Implementation.....	18
Reflection.....	18
Task 3.....	19
IP Division.....	19
1. Working operation.....	20
2. IP Division Table.....	22
3. Details Range IP Address.....	23
4. IP Allocation.....	24
Reflection.....	26
Task 4.....	27
Topology.....	27
EIGRP Route.....	28
Static Route.....	29
Configuration.....	30

Routing Table.....	36
End-to-End Connection.....	39
Reflection.....	42
Task 5.....	43
Topology.....	43
Subnetting.....	44
Calculation operation.....	45
Details Range IP Address.....	46
IP Address assignment.....	48
Configuration.....	49
VLAN Table.....	55
End-to-End Connection.....	57
Reflection.....	61
Task 6.....	62
DHCP Configuration.....	62
Security Configuration.....	70
DHCP Message Exchange Process.....	70
DHCP Offer.....	72
DHCP Request.....	74
DHCP Acknowledge.....	75
End-to-End Connection.....	77
Reflection.....	78
Conclusion.....	79
Team Members and Responsibilities.....	80
References.....	82
Appendices.....	83

Table of Figures

Figure 1.1: Floor 1.....	8
Figure 1.2: Floor 2.....	9
Figure 1.3: General Purpose Lab.....	10
Figure 1.4: Security Lab.....	11
Figure 1.5: IoT Lab.....	12
Figure 1.6: Network Lab.....	13
Figure 1.7: Video Conference Room.....	14
Figure 2.1: Network topology diagram in packet tracer.....	16
Figure 3.1: Subnetting Working Operation (Part 1).....	19
Figure 3.2: Subnetting Working Operation (Part 2).....	20
Figure 4.1: Faculty building topology in packet tracer.....	26
Figure 4.2: EIGRP Route highlight.....	27
Figure 4.3: Static Route highlight.....	28
Figure 4.4: Core router routing table.....	35
Figure 4.5: GPL router routing table.....	35
Figure 4.6: SL router routing table.....	36
Figure 4.7: NL router routing table.....	36
Figure 4.8: IOT router routing table.....	37
Figure 4.10: PC 0 to PC 16 ping.....	39
Figure 4.11: PC 0 to PC 4 ping.....	39
Figure 4.12: PC 0 to PC 8 ping.....	40
Figure 4.13: PC 0 to PC 18 ping.....	40
Figure 4.14: PC 0 to PC 12 ping.....	40
Figure 4.15: PC 0 to Server ping.....	41
Figure 5.1: Faculty building topology in packet tracer.....	42
Figure 5.2: Subnetting Calculation.....	44
Figure 5.3: Staff Room 1 (VLAN 1) table.....	54

Figure 5.4: Staff Room 2 (VLAN 2) table.....	54
Figure 5.5: Staff Room 3 (VLAN 3) table.....	54
Figure 5.6: Staff Room 4 (VLAN 4) table.....	55
Figure 5.7: Ping to PC1 test.....	56
Figure 5.8: Ping to PC16 test.....	57
Figure 5.9: Ping to PC4.....	57
Figure 5.10: Ping to PC8.....	57
Figure 5.11: Ping to PC18.....	58
Figure 5.12: Ping to PC12.....	58
Figure 5.13: Ping to Server.....	58
Figure 5.14: Ping to PC20.....	59
Figure 5.15: Ping to PC22.....	59
Figure 5.16: Ping to PC24.....	59
Figure 5.17: Ping to PC6.....	60
Figure 6.1: General Purpose Lab DHCP configuration.....	61
Figure 6.2: Video Conference Room 1 DHCP configuration.....	62
Figure 6.3: IoT Lab DHCP configuration.....	63
Figure 6.4: Video Conference Room 2 DHCP configuration.....	63
Figure 6.5: Computer Security Lab DHCP configuration.....	64
Figure 6.6: Network Lab DHCP configuration.....	65
Figure 6.7: Staff Room 1 DHCP configuration.....	66
Figure 6.8: Staff Room 2 DHCP configuration.....	66
Figure 6.9: Staff Room 3 DHCP configuration.....	67
Figure 6.10 : Staff Room 4 DHCP configuration.....	68
Figure 6.11: DHCP client send discover packet.....	70
Figure 6.12: Relay agent send to helper address.....	70
Figure 6.13: DHCP server sends offer packet.....	71
Figure 6.14: Relay agent forward reply to DHCP client.....	72
Figure 6.15: DHCP client sends out a request packet.....	72
Figure 6.16: Relay agent forwards request packet to DHCP server.....	73

Figure 6.17: The DHCP server sends an acknowledgment packet.....	74
Figure 6.18: DHCP client receives acknowledgment packet.....	75
Figure 6.19: DHCP client sends IP address.....	75
Figure 6.20: PDU list.....	76

Introduction

Basically, we do network operations virtualization for the Faculty of Computing. We provide a floor plan for a two-storey building that includes General purpose labs, Security lab, IoT lab, Network Lab and Video Conference Room. Next, we set up the network topology diagram that can help us to implement the network. In simple terms, we provide a building and the network topology to make sure that all the users can use the facilities and networking capabilities.

Our objective is to design and implement a network infrastructure that meets the current needs of the new building while being scalable for future expansion. Besides that, we aim to create a state-of-the-art learning environment that supports advanced research and practical training in various technological fields. Moreover, we also want to ensure robust and reliable network connectivity across all labs and conferencing rooms to support uninterrupted workflow and collaboration.

We have assumptions that the building's physical infrastructure will be enabled to support the installation of necessary networking hardware. Then, enough power supply and reliable backup system to ensure continuous operation and we also have sufficient budget allocated for high quality hardware and software. Security measures will be implemented to protect the network from any threats and unauthorized access.

Project Background

In response to the growing needs of our organization, the decision was made to construct a new two-storey building dedicated to supporting various technological and research activities. This new infrastructure aims to enhance the organization's capabilities by housing four specialized labs and two video conferencing rooms designed for virtual project meetings. The building will facilitate cutting-edge research, practical training, and seamless virtual collaboration. The four labs are strategically designed to cater to different areas of expertise:

1. General Purpose Lab: Equipped with 31 workstations to support a wide range of computing activities.
2. Computer Security Lab: Featuring 25 workstations, this lab is tailored for cybersecurity training and research.
3. Network Lab: With 32 workstations, this lab is dedicated to networking studies and practicals.
4. IoT Lab: Housing 25 workstations, the IoT lab is focused on the burgeoning field of Internet of Things technologies.

Meanwhile, the video conferencing rooms, each equipped with 2 workstations, are intended to support remote collaboration and virtual meetings, facilitating efficient communication and project management.

Our objective is to design and implement a network infrastructure that meets the current needs of the new building while being scalable for future expansion. Besides that, we aim to create a state-of-the-art learning environment that supports advanced research and practical training in various technological fields. Moreover, we also want to ensure robust and reliable network connectivity across all labs and conferencing rooms to support uninterrupted workflow and collaboration.

The project relied heavily on the Cisco Packet Tracer for designing and simulating the network topology. This tool allowed us to visualize and configure the network components, ensuring a practical and functional design before actual implementation.

For this project, there are several key considerations that we prioritised. One of it is that the network design had to be scalable to support future growth and additional devices. Besides that, implementing robust security measures to protect sensitive data and ensure secure communication within the network. Moreover, ensuring high performance and low latency to support the demanding requirements of research and practical training. This project not only addresses the immediate networking needs of the new building but also sets a foundation for future technological advancements and expansions. By creating a well-designed and efficiently managed network infrastructure, we aim to significantly enhance the operational capabilities and educational experience within the organization.

Task 1

Floor Plan: Floor 1

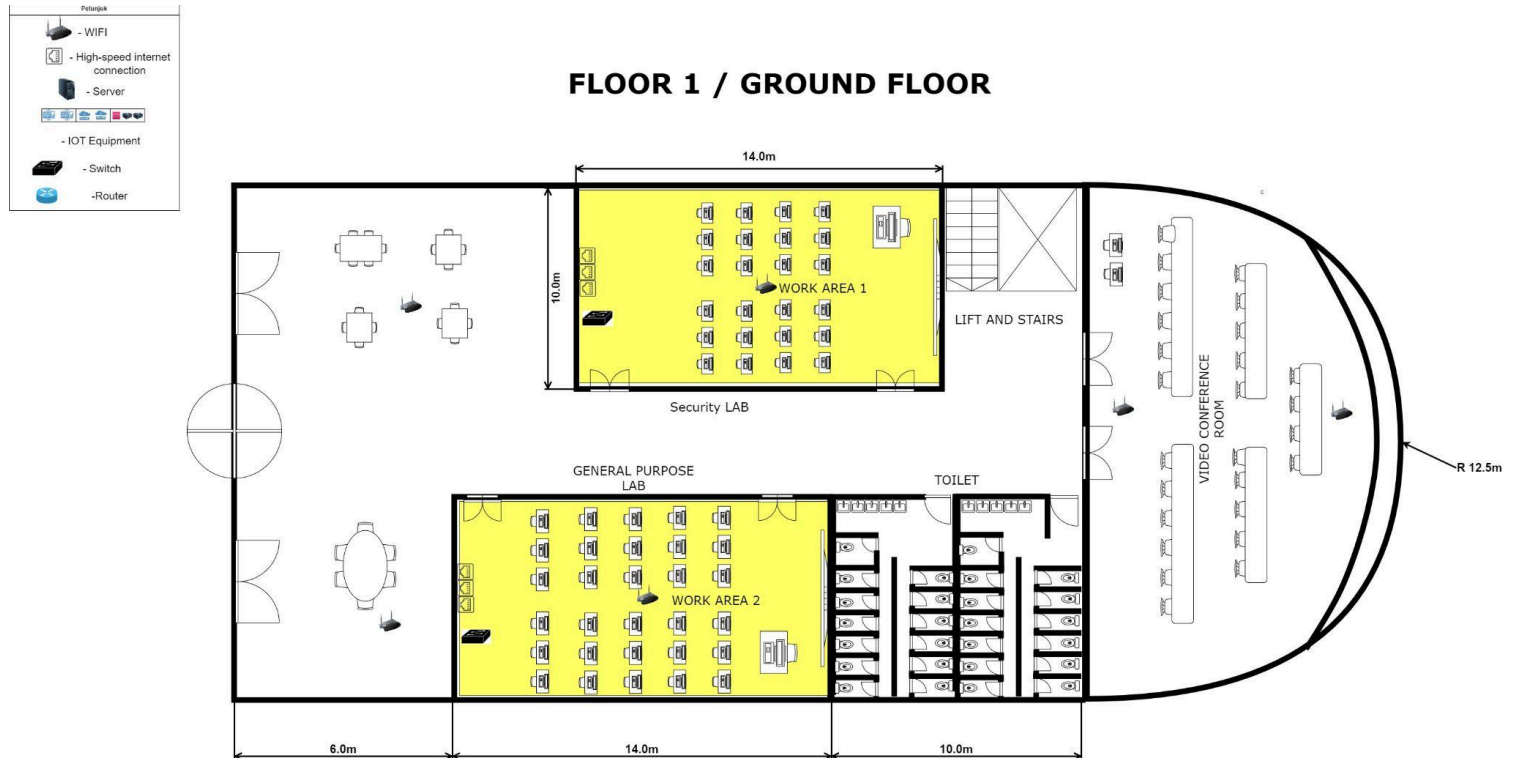


Figure 1.1: Floor 1

Scale:

1 m = 1 m

Floor Plan: Floor 2

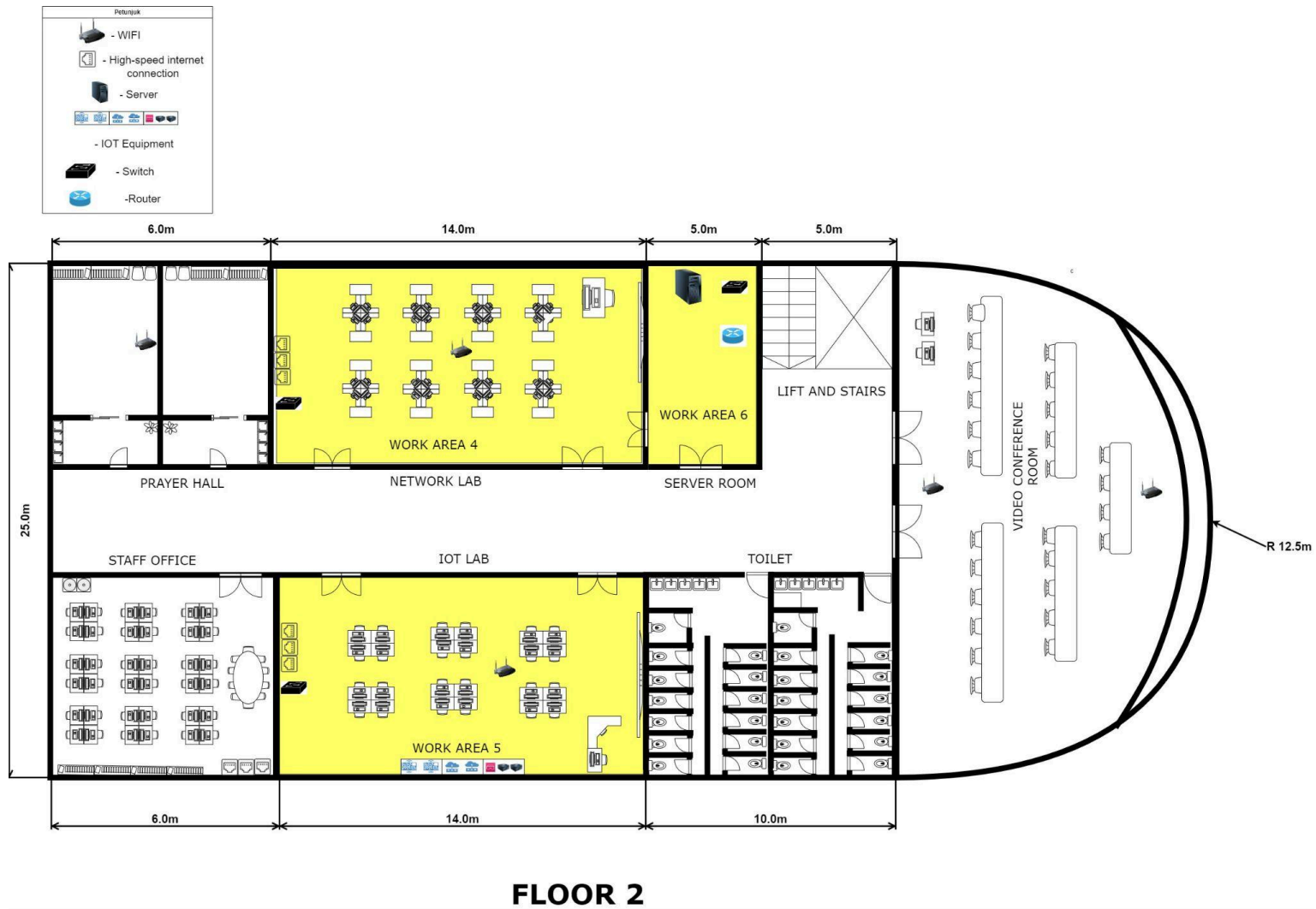


Figure 1.2: Floor 2

Scale:

1 m = 1 m

Plan Details: General Purpose Lab

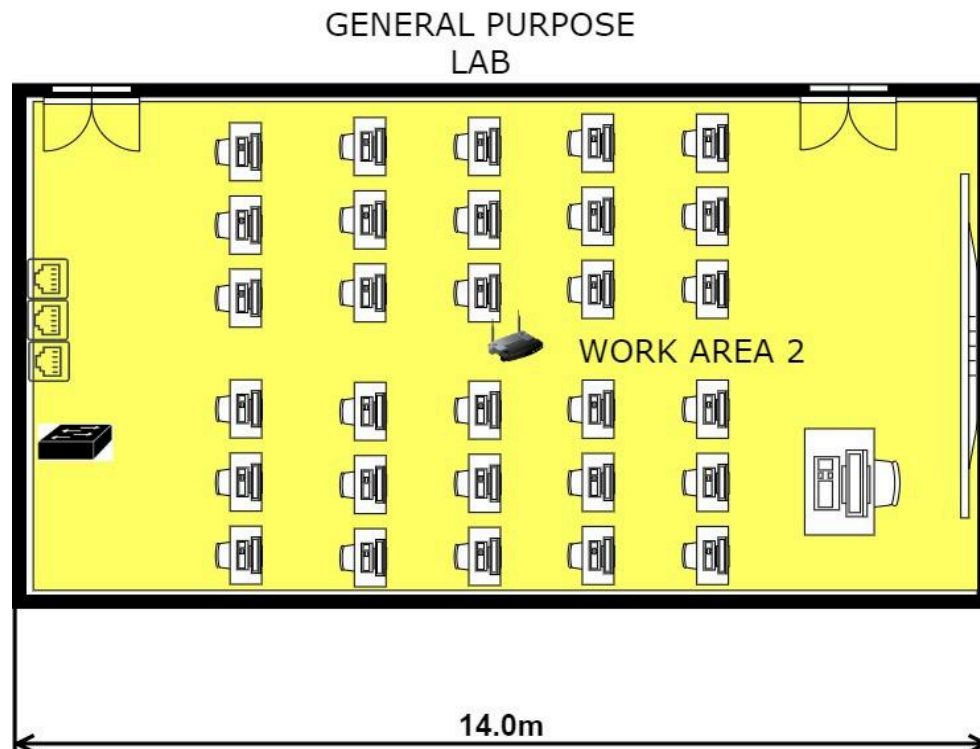


Figure 1.3: General Purpose Lab

A general-purpose lab is designed to accommodate a wide range of activities. It can be used for various types of work. The layout is very crucial to allow easy access to any presentation or any movement of lecturers and students. Safety is at the top of our list. We designed appropriate safety measures in place for our lab, including proper ventilation. This lab is equipped with computers, a presentation board, and a switch. The purpose of this lab is used for teaching and learning. Other than that, for the research purposes. This could involve developing new software or constructing new algorithms.

Plan Details: Security Lab

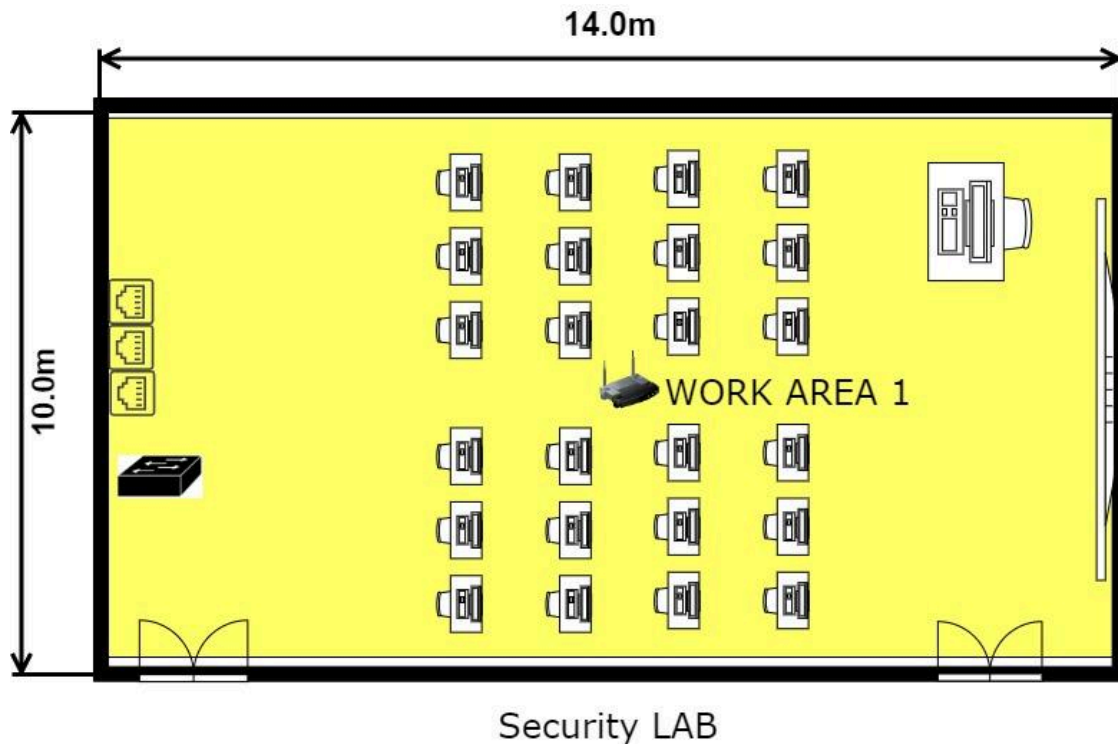


Figure 1.4: Security Lab

This lab is designed to help students with their security work. The lab is equipped with the latest tools and technologies to study various aspects, especially computer security, and network security. This lab has infrastructure that includes high-speed internet, secure servers, and multiple high-quality workstations with the latest hardware and software. We set up the lab environment similar to the real-world scenario, allowing students to study and apply attack strategies. For safety measures, we ensure that all the activities that happen in the lab do not negatively impact the rest of the campus network. Other than that, we also prepared with a few upgrading and updating steps to ensure the system is updated with the latest software, and technologies and safe from any threats.

Plan Details: IoT Lab

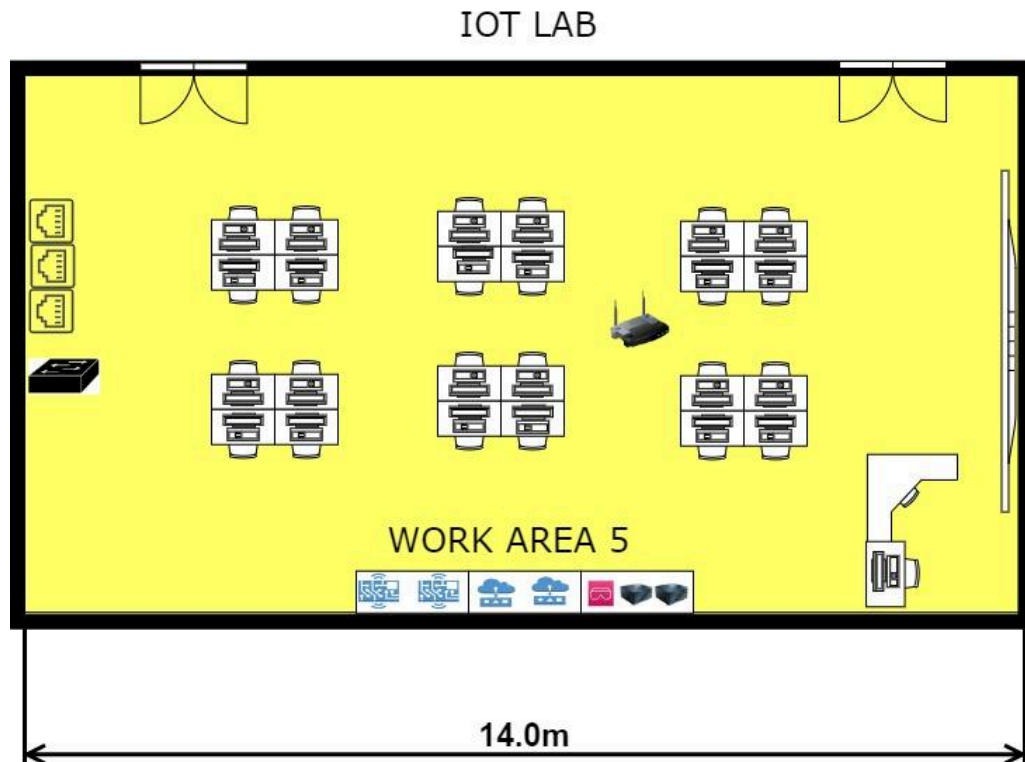


Figure 1.5: IoT Lab

The IoT Lab is a specialized lab designed to help students get hands-on experience on real-world IoT devices, and help to raise innovative ideas and skill development among students. Workstations are arranged in a grid style to facilitate better and more productive discussions between students. With high-quality IoT equipment equipped in the lab, students can better understand the potential of IoT technology as well as generate ideas for the research and development of IoT devices, students can also explore for themselves the IoT ecosystems that are becoming more commonly used in the real world. This lab will help students create, test, and do experiments on IoT devices, preparing them for future careers in this field.

Plan Details: Network Lab

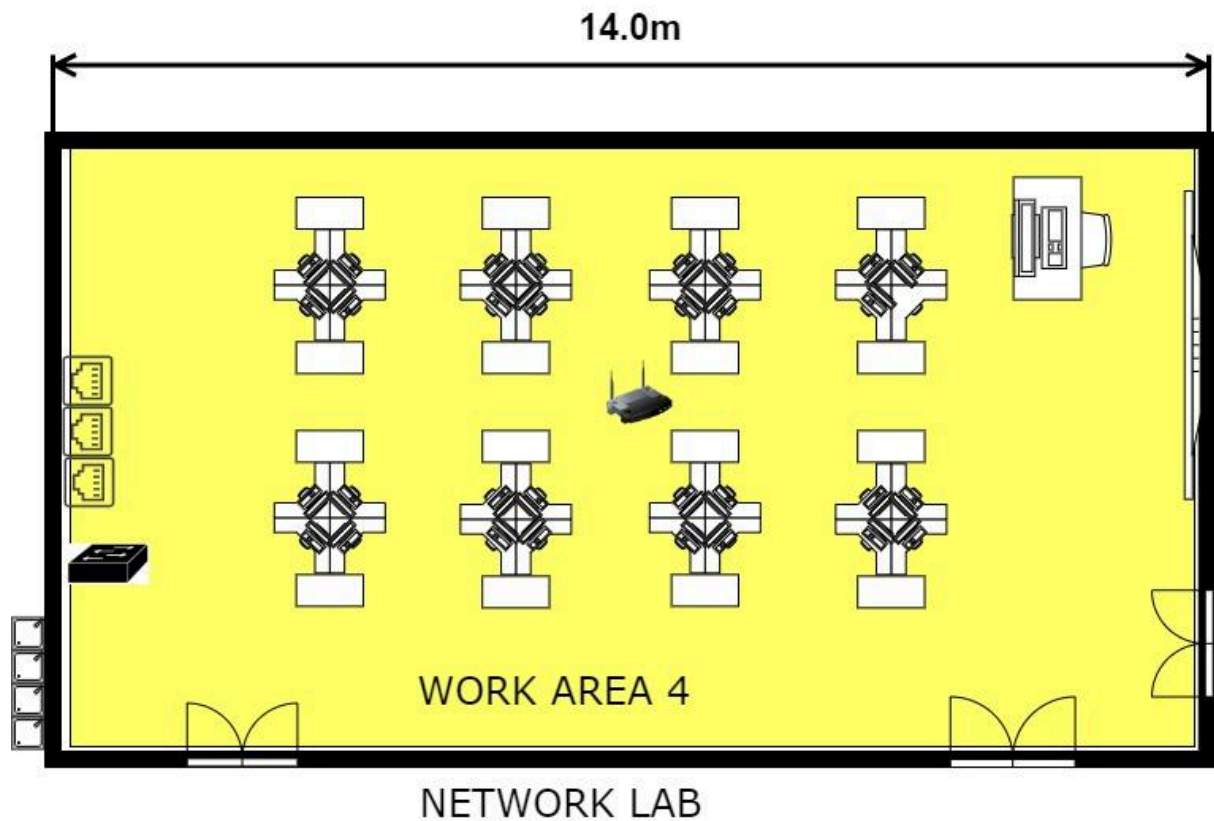


Figure 1.6: Network Lab

The Cisco lab is designed to help students build, test, and refine their skills in handling network infrastructures. With the workstations arranged in a comfortable “X” arrangement, this enables easier communication and discussion between students. The lab is equipped with the needed networking hardware and software to ensure a high-quality experience in using networking equipment among students. This lab can be used for research and development, certification preparation, and training purposes. High-quality routers and high-speed networks will ensure the best possible networking environment for students to work in.

Plan Details: Video Conference Room

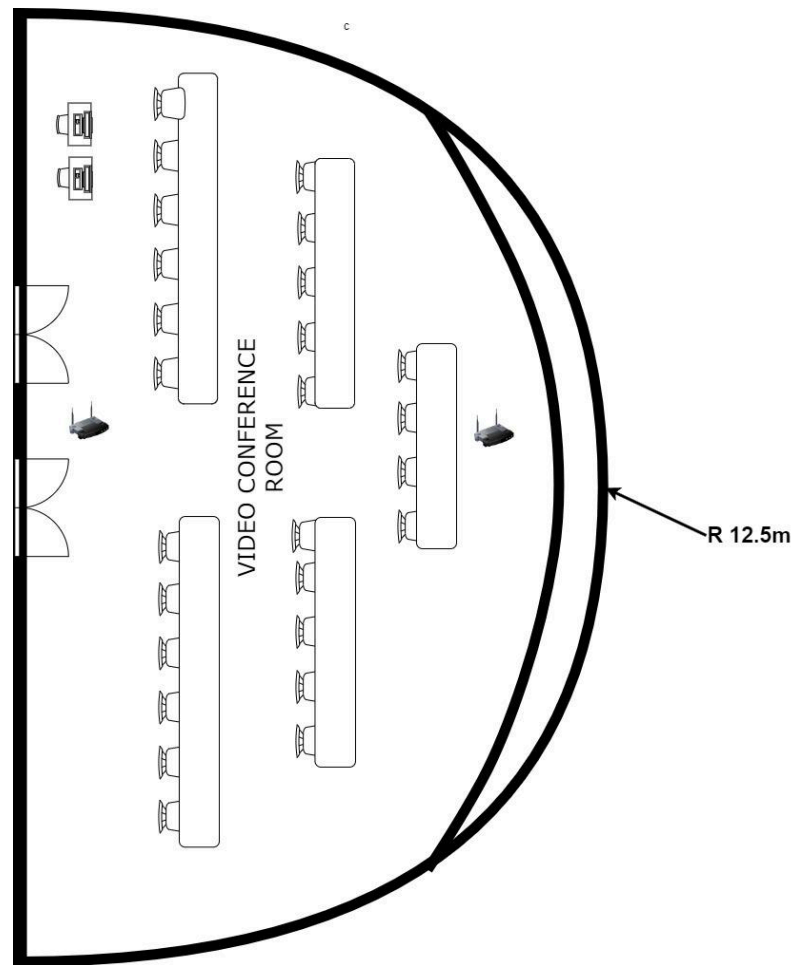


Figure 1.7: Video Conference Room

The video conferencing room is designed to resemble a cinema hall, with a large smart display that can connect to WiFi to showcase online meetings or presentation slides. This room has two workstations and two access points for seamless connectivity. Given its size, the video conferencing room can also be utilised for hosting events, with a stage set up at the front. The Internet connectivity is robust and fast, ensuring that there are no buffering issues or distractions during conferences. It's worth mentioning that both the video conferencing room and floors 1 and 2 have the same design.

Reflection

This task emphasised the importance of teamwork and collaboration. It was crucial to select team members who could contribute effectively to the project. We learned how to coordinate and communicate effectively as a team, which is essential for successful project management. Additionally, the task allowed us to understand the significance of a well-thought-out project plan and the ability to adapt and make changes as needed.

Task 2

Topology Design

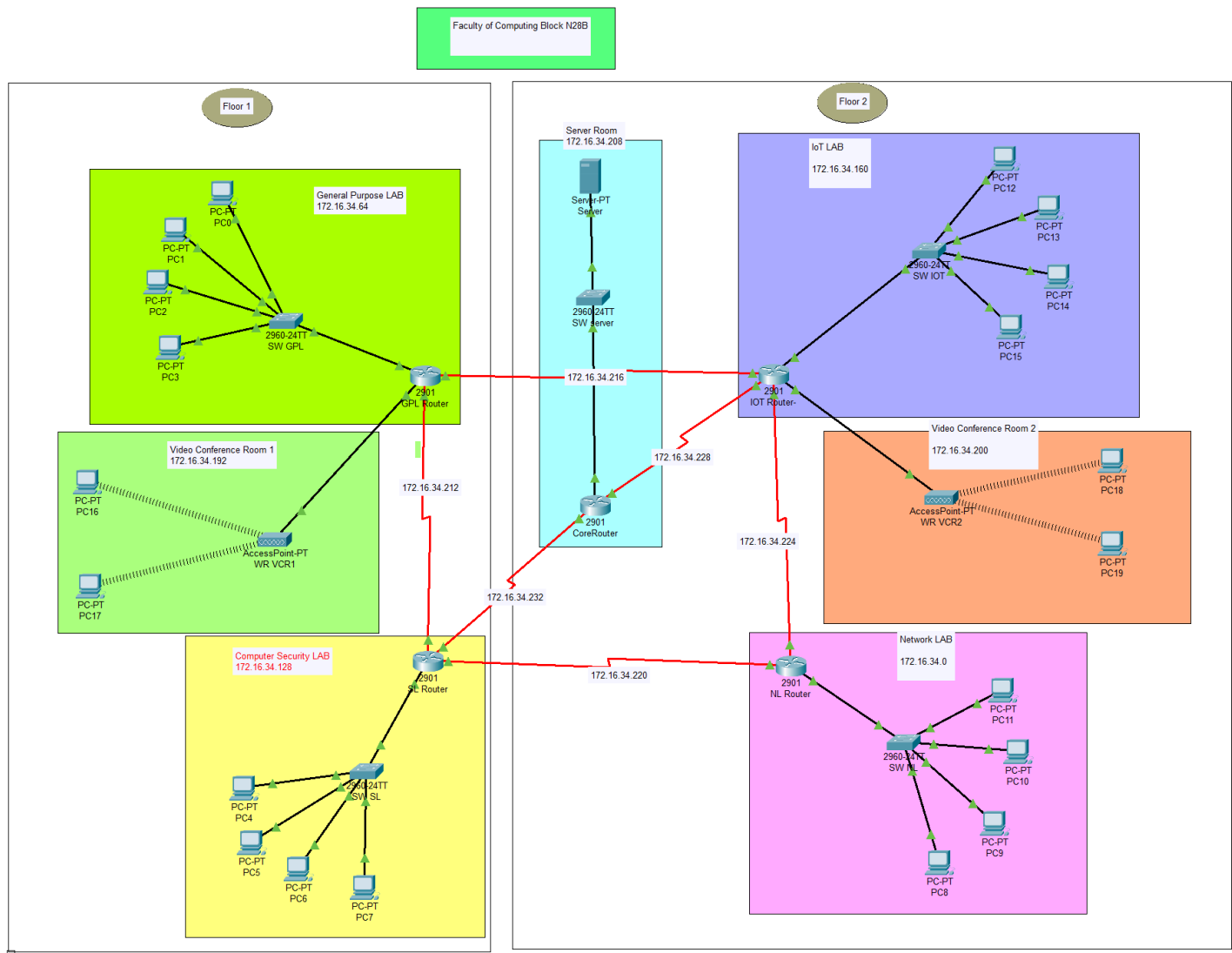


Figure 2.1: Network topology diagram in packet tracer

Implementation

In this topology network, we applied a two-tier hierarchical network design and combined it with a partial mesh topology, which divides the network into two parts: core and access layer [4]. Each layer has its task.

The routers in the topology are almost all directly connected to each other and offer several advantages. These direct connections create redundancy, which ensures that if one path fails due to a link or router failure, data transmission can automatically reroute through an alternate path. This fault tolerance is crucial for maintaining uninterrupted communication. Additionally, multiple direct connections enhance the robustness of the network. Robustness refers to the network's ability to withstand failures or disruptions without significant impact. With direct links, even if individual connections experience issues, the overall network remains reliable. Furthermore, direct connections allow for load balancing. By distributing network traffic across available paths, congestion on any single link is prevented, leading to better performance and efficient resource utilization [3].

In the core layer, we have added one core router to provide high-speed data forwarding across various network segments. Since the core aggregates traffic from all devices in the distribution layer, it needs to be redundant and highly available [4].

Lastly, in the access layer, we add a local network with labs and video conference rooms which each is a small LAN. This LAN contains an end device such as a PC and has an intermediary device such as a layer 2 switch for switching purposes and an access point for wireless connection.

Reflection

This task provided hands-on experience with network simulation tools. We gained a deeper understanding of network topology design and the practical application of networking concepts. The task also highlighted the importance of attention to detail and accuracy in setting up network devices to ensure a functional and efficient network.

Task 3

IP Division

To ensure there is no wasted IP address we will do a variable length subnet mask (VLSM). The network address will be subnetted and then re-subnetted multiple times until all areas in Table 1 have their subnet. The network address we will be using is 172.16.34.0/23. Notes in for WAN the first router in the name of the WAN will be assigned with the first usable IP address.

Area	Number of Hosts
Video conferencing rooms 1	3
Video conferencing rooms 2	3
General purpose lab	32
Computer Security lab	26
Network lab	33
IoT lab	26
Server Room	2

Table 1: Number of Hosts for Each Area

WAN	Number of Hosts
WAN GPL - SL	2
WAN GPL - IOT	2
WAN SL - NL	2
WAN NL - IOT	2
WAN SR - IOT	2
WAN SR - SL	2

Table 2: Number of Hosts for Each WAN

1. Working operation

172	16	34	0
10101100.00010000.00100010.00000000			
	NP	HP	
N=33	CS=10T=26	S=WAN=2	
$2^x = 33$	$2^x = 26$	$2^x = 2$	
$x = 6$	$x = 5$	$x = 2$	
GP=32	VC1=VC2=3		
$2^x = 32$	$2^x = 3$		
$x = 6$	$x = 3$		
10101100.00010000.00100010.00000000 [0]			
$S_0 = N$	.00	111111	[63]
	.01	000000	[64]
$S_1 = GP$	.01	111111	[127]
	.100	000000	[128]
$S_2 = CS$	.100	111111	[159]
	.101	000000	[160]
$S_3 = 10T$	.101	111111	[191]
	.11000	000	[192]
$S_4 = VC1$	.11000	111	[199]
	.11001	000	[200]
$S_5 = VC2$	.11001	111	[207]
	.110100	00	[208]
$S_6 = S$	.110100	11	[211]
	.110101	00	[212]

Figure 3.1: Subnetting Working Operation (Part 1)

10101100.00001000.00100010.110101	00[212]
$S_7 = \text{WAN GPL-SL}$	.11010111[215]
	.11011000[216]
$S_8 = \text{WAN GPL-IOT}$	.11011011[219]
	.11011100[220]
$S_9 = \text{WAN SL-NL}$	.11011111[223]
	.11100000[224]
$S_{10} = \text{WAN NL-IOT}$	.11100011[227]
	.11100100[228]
$S_{11} = \text{WAN SR-IOT}$	.11100111[231]
	.11101000[232]
$S_{12} = \text{WAN SR-SL}$	.11101011[235]
	.11101100[236]

Figure 3.2: Subnetting Working Operation (Part 2)

2. IP Division Table

Subnet	Area	IP Address (Binary)	IP Address (Decimal)	Subnet Mask
0	Network lab	10101100.00010000.00100010.00000000	172.16.34.0	/26
1	General Purpose lab	10101100.00010000.00100010.01000000	172.16.34.64	/26
2	Computer Security lab	10101100.00010000.00100010.10000000	172.16.34.128	/27
3	IoT lab	10101100.00010000.00100010.10100000	172.16.34.160	/27
4	Video conferencing rooms 1	10101100.00010000.00100010.11000000	172.16.34.192	/29
5	Video conferencing rooms 2	10101100.00010000.00100010.11001000	172.16.34.200	/29
6	Server room	10101100.00010000.00100010.11010000	172.16.34.208	/30
7	WAN GPL - SL	10101100.00010000.00100010.11010100	172.16.34.212	/30
8	WAN GPL - IOT	10101100.00010000.00100010.11011000	172.16.34.216	/30
9	WAN SL - NL	10101100.00010000.00100010.11011100	172.16.34.220	/30
10	WAN NL - IOT	10101100.00010000.00100010.11100000	172.16.34.224	/30
11	WAN SR - IOT	10101100.00010000.00100010.11100100	172.16.34.228	/30
12	WAN SR - SL	10101100.00010000.00100010.11101000	172.16.34.232	/30

Table 3: IP Division table

3. Details Range IP Address

Subnet	Area	Network Address	Broadcast Address	Usable Range IP Address
0	Network lab	172.16.34.0	172.16.34.63	172.16.34.1-172.16.34.62
1	General Purpose lab	172.16.34.64	172.16.34.127	172.16.34.65-172.16.34.126
2	Computer Security lab	172.16.34.128	172.16.34.159	172.16.34.129-172.16.34.158
3	IoT lab	172.16.34.160	172.16.34.191	172.16.34.161-172.16.34.190
4	Video conferencing rooms 1	172.16.34.192	172.16.34.199	172.16.34.193-172.16.34.198
5	Video conferencing rooms 2	172.16.34.200	172.16.34.207	172.16.34.201-172.16.34.206
6	Server room	172.16.34.208	172.16.34.211	172.16.34.209 - 172.16.34.210
7	WAN GPL - SL	172.16.34.212	172.16.34.215	172.16.34.213 - 172.16.34.214
8	WAN GPL - IOT	172.16.34.216	172.16.34.219	172.16.34.217 - 172.16.34.218
9	WAN SL - NL	172.16.34.220	172.16.34.223	172.16.34.221 - 172.16.34.222
10	WAN NL - IOT	172.16.34.224	172.16.34.227	172.16.34.225 - 172.16.34.226
11	WAN SR - IOT	172.16.34.228	172.16.34.231	172.16.34.229 - 172.16.34.230
12	WAN SR - SL	172.16.34.232	172.16.34.235	172.16.34.233 - 172.16.34.234

Table 4: Details Range IP Address

4. IP Allocation

- Network Lab

Device	Interface	IP Address	Gateway Address
PC8	Fa0	172.16.34.1	172.16.34.62
PC9	Fa0	172.16.34.2	172.16.34.62
PC10	Fa0	172.16.34.3	172.16.34.62
PC11	Fa0	172.16.34.4	172.16.34.62

Table 5: Network Lab IP allocation

- General Purpose Lab

Device	Interface	IP Address	Gateway Address
PC0	Fa0	172.16.34.65	172.16.34.126
PC1	Fa0	172.16.34.66	172.16.34.126
PC2	Fa0	172.16.34.67	172.16.34.126
PC3	Fa0	172.16.34.68	172.16.34.126

Table 6: General Purpose Lab IP allocation

- Computer Security Lab

Device	Interface	IP Address	Gateway Address
PC4	Fa0	172.16.34.129	172.16.34.158
PC5	Fa0	172.16.34.130	172.16.34.158
PC6	Fa0	172.16.34.131	172.16.34.158
PC7	Fa0	172.16.34.132	172.16.34.158

Table 7: Computer Security Lab IP allocation

- Iot Lab

Device	Interface	IP Address	Gateway Address
PC12	Fa0	172.16.34.161	172.16.34.190
PC13	Fa0	172.16.34.162	172.16.34.190
PC14	Fa0	172.16.34.163	172.16.34.190
PC15	Fa0	172.16.34.164	172.16.34.190

Table 8: IoT Lab IP allocation

- Video conferencing room 1

Device	Interface	IP Address	Gateway Address
PC16	Wireless	DHCP	172.16.34.198
PC17	Wireless	DHCP	172.16.34.198

Table 9: Video Conference Room 1 IP allocation

- Video conference room 2

Device	Interface	IP Address	Gateway Address
PC18	Wireless	DHCP	172.16.34.206
PC19	Wireless	DHCP	172.16.34.206

Table 10: Video Conference Room 2 IP allocation

- Server Room

Device	Interface	IP Address	Gateway Address
Server	Fa0	172.16.34.209	172.16.34.214

Table 11: Server Room IP allocation

Reflection

This task enhanced our understanding of IP addressing and subnetting. We learned the importance of proper IP address allocation and the potential issues that can arise from IP address conflicts. The task also reinforced our skills in manual subnetting and calculating network addresses, which are critical for effective network management.

Task 4

Topology

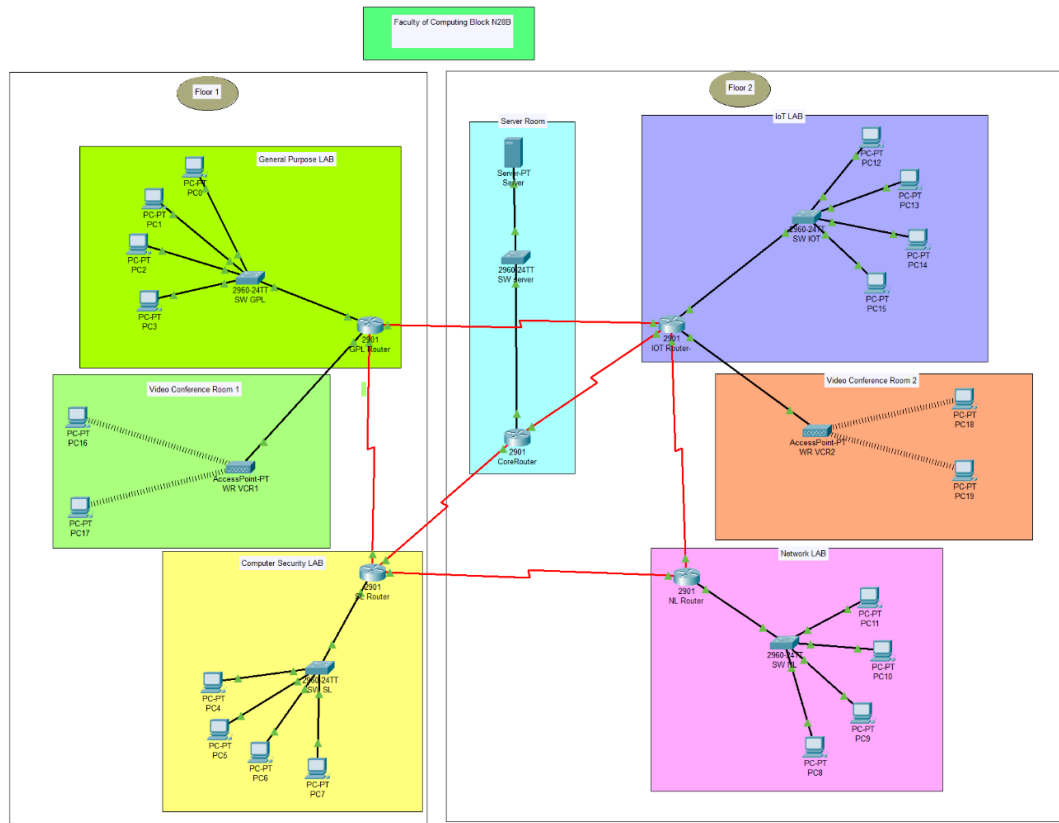


Figure 4.1: Faculty building topology in packet tracer

The topology diagram consists of 5 routers, 2 access points, and Local Area Networks (LANs). To ensure end-to-end connectivity we implement dynamic routing by using Enhanced Interior Gateway Routing Protocol (EIGRP). The autonomous system (AS) is set as 10. An AS is a collection of IP networks and routers under the control of one entity that presents a common routing policy to the Internet [1]. To ensure that all routers belong to the same network and can communicate correctly with each other, all routers have the same AS. EIGRP is called a hybrid or advanced distance vector protocol, and most of the rules that apply to RIP also apply here [2]. Besides dynamic routing, we also implement static routing routes to the server which is an important service. All routers except core_router were

configured with equal-cost static routes for the alternative route with metric 6 to the server network. The routers have multiple paths to reach the server network, and these paths have the same cost, providing redundancy and load balancing.

EIGRP Route

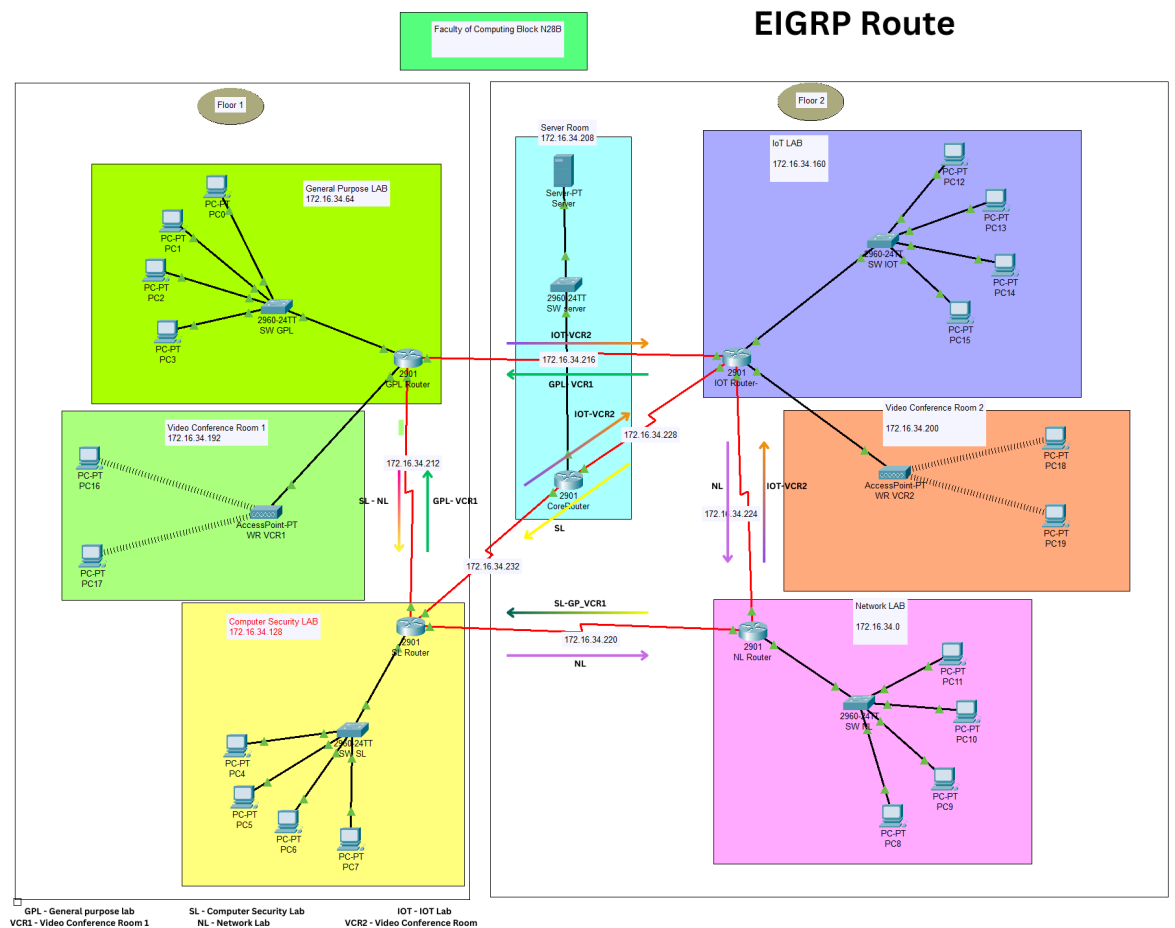


Figure 4.2: EIGRP Route highlight

Static Route

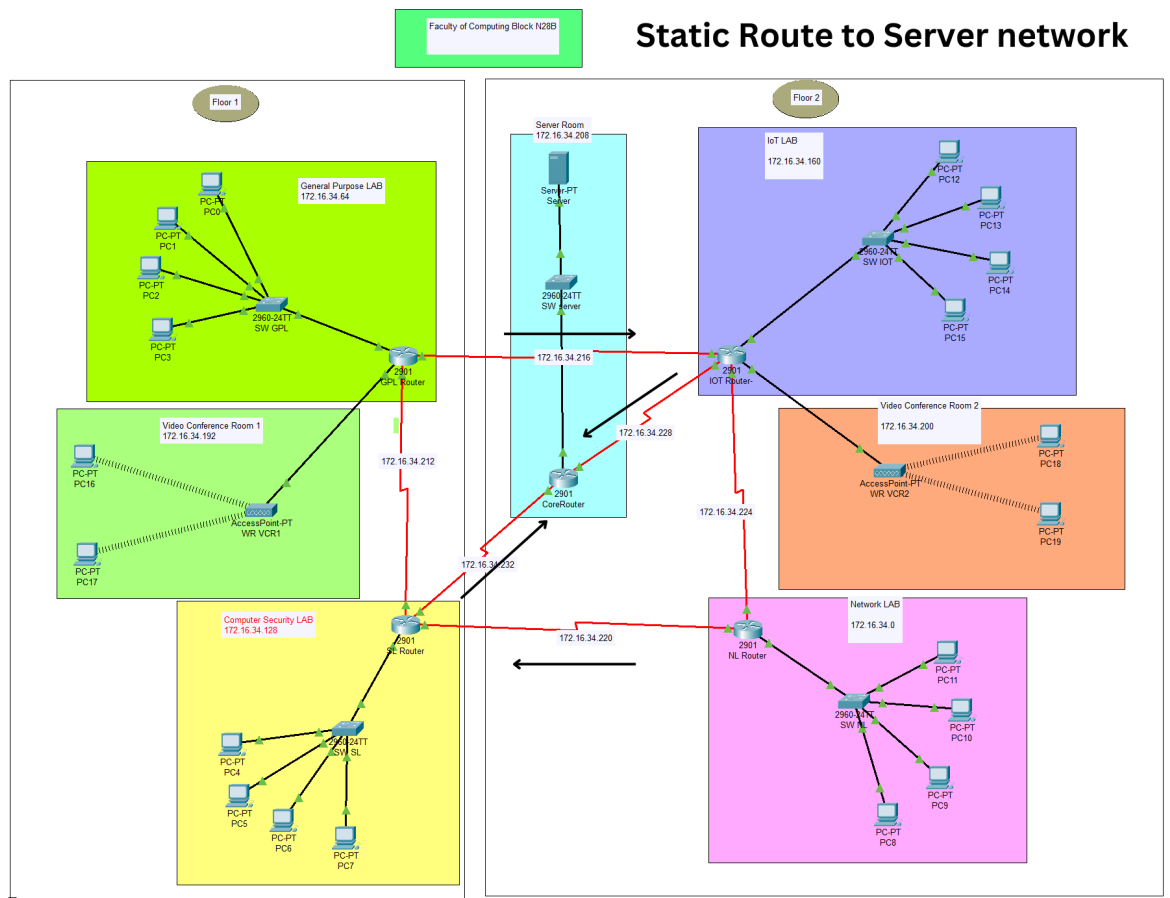


Figure 4.3: Static Route highlight

Configuration

Core Router

```
Unset
Enable
Configuration terminal
hostname Core_Router
interface serial 0/1/0
ip address 172.16.34.229 255.255.255.252
clock rate 64000
no shutdown
interface serial 0/1/1
ip address 172.16.34.233 255.255.255.252
clock rate 64000
no shutdown
interface gigabitEthernet 0/0
ip address 172.16.34.210 255.255.255.252
no shutdown
Router eigrp 10
network 172.16.34.228 255.255.255.252
network 172.16.34.232 255.255.255.252
no auto-summary
exit
```

SW_Server

```
Unset
enable
configure terminal
ip default-gateway 172.16.34.210
```

GPL_Router

```
Unset
enable
configuration terminal
hostname GPL_Router
interface serial 0/0/1
ip address 172.16.34.213 255.255.255.252
clock rate 64000
no shutdown
interface serial 0/0/0
ip address 172.16.34.217 255.255.255.252
clock rate 64000
no shutdown
interface gigabitEthernet 0/1
ip address 172.16.34.126 255.255.255.192
no shutdown
interface gigabitEthernet 0/0
ip address 172.16.34.198 255.255.255.248
no shutdown
exit
ip dhcp pool VCR1
network 172.16.34.192 255.255.255.248
default-router 172.16.34.198
exit
ip route 172.16.34.208 255.255.255.252 serial 0/0/0 6
ip route 172.16.34.208 255.255.255.252 serial 0/0/1
router eigrp 10
network 172.16.34.64 255.255.255.192
network 172.16.34.192 255.255.255.248
network 172.16.34.212 255.255.255.252
network 172.16.34.216 255.255.255.252
no auto-summary
exit
```


SW_GPL

```
Unset
enable
configuration terminal
ip default-gateway 172.16.34.126
```

SL_Router

```
Unset
enable
configure terminal
hostname SL_Router
interface serial 0/0/0
ip address 172.16.34.218 255.255.255.252
no shutdown
interface serial 0/0/1
ip address 172.16.34.221 255.255.255.252
clock rate 64000
no shutdown
interface serial 0/1/0
ip address 172.16.34.230 255.255.255.252
no shutdown
interface gigabitEthernet 0/0
ip address 172.16.34.158 255.255.255.224
no shutdown
exit
ip route 172.16.34.208 255.255.255.252 serial 0/1/0
ip route 172.16.34.208 255.255.255.252 serial 0/0/1 6
ip route 172.16.34.208 255.255.255.252 serial 0/0/0 6
router eigrp 10
network 172.16.34.128 255.255.255.224
network 172.16.34.220 255.255.255.252
network 172.16.34.228 255.255.255.252
no auto-summary
exit
```

SW_SL

```
Unset
enable
configure terminal
hostname SW_SL
ip default-gateway 172.16.34.158
```

NL_Router

```
Unset
enable
configure terminal
hostname NL_Router
interface serial 0/0/1
ip address 172.16.34.222 255.255.255.252
no shutdown
interface serial 0/0/0
ip address 172.16.34.225 255.255.255.252
clock rate 64000
no shutdown
interface gigethernet 0/0
ip address 172.16.34.62 255.255.255.192
no shutdown
exit
ip route 172.16.34.208 255.255.255.252 serial0/0/1
ip route 172.16.34.208 255.255.255.252 serial 0/0/0 6
router eigrp 10
network 172.16.34.0 255.255.255.192
network 172.16.34.224 255.255.255.252
network 172.16.34.220 255.255.255.252
no auto-summary
exit
```

SW_NL

```
Unset
enable
configure terminal
ip default-gateway 172.16.34.62
```

IOT_Router

```
Unset
enable
configure terminal
hostname IOT_Router
interface serial 0/0/0
ip address 172.16.34.226 255.255.255.252
no shutdown
interface serial 0/0/1
ip address 172.16.34.214 255.255.255.252
no shutdown
interface serial 0/1/1
ip address 172.16.34.234 255.255.255.252
no shutdown
interface gigabitEthernet 0/1
ip address 172.16.34.190 255.255.255.224
no shutdown
interface gigabitEthernet 0/0
ip address 172.16.34.206 255.255.255.248
no shutdown
exit
ip dhcp pool VCR2
network 172.16.34.200 255.255.255.248
default-router 172.16.34.206
exit
ip route 172.16.34.208 255.255.255.252 serial 0/1/1
ip route 172.16.34.208 255.255.255.252 serial 0/0/1 6
```

```
ip route 172.16.34.208 255.255.255.252 serial 0/0/0 6
router eigrp 10
network 172.16.34.160 255.255.255.224
network 172.16.34.200 255.255.255.248
network 172.16.34.212 255.255.255.252
network 172.16.34.224 255.255.255.252
network 172.16.34.232 255.255.255.252
no auto-summary
exit
```

SW_IOT

```
Unset
enable
configure terminal
hostname SW_IOT
ip default-gateway 172.16.34.190
```

Routing Table

```
Core_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 16 subnets, 5 masks
D       172.16.34.0/26 [90/2682112] via 172.16.34.234, 00:26:57, Serial0/1/1
        [90/2682112] via 172.16.34.230, 00:26:53, Serial0/1/0
D       172.16.34.64/26 [90/2682112] via 172.16.34.230, 00:26:55, Serial0/1/0
        [90/2682112] via 172.16.34.234, 00:26:53, Serial0/1/1
D       172.16.34.128/27 [90/2170112] via 172.16.34.230, 00:26:57, Serial0/1/0
D       172.16.34.160/27 [90/2170112] via 172.16.34.234, 00:26:57, Serial0/1/1
D       172.16.34.192/29 [90/2682112] via 172.16.34.230, 00:26:55, Serial0/1/0
        [90/2682112] via 172.16.34.234, 00:26:53, Serial0/1/1
D       172.16.34.200/29 [90/2170112] via 172.16.34.234, 00:26:57, Serial0/1/1
C       172.16.34.208/30 is directly connected, GigabitEthernet0/0
L       172.16.34.210/32 is directly connected, GigabitEthernet0/0
D       172.16.34.212/30 [90/2681856] via 172.16.34.234, 00:26:53, Serial0/1/1
D       172.16.34.216/30 [90/2681856] via 172.16.34.230, 00:26:55, Serial0/1/0
D       172.16.34.220/30 [90/2681856] via 172.16.34.230, 00:26:57, Serial0/1/0
D       172.16.34.224/30 [90/2681856] via 172.16.34.234, 00:26:57, Serial0/1/1
C       172.16.34.228/30 is directly connected, Serial0/1/0
L       172.16.34.229/32 is directly connected, Serial0/1/0
C       172.16.34.232/30 is directly connected, Serial0/1/1
L       172.16.34.233/32 is directly connected, Serial0/1/1

Core_Router#
```

Figure 4.4: Core router routing table

```
GPL_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 17 subnets, 5 masks
D       172.16.34.0/26 [90/2682112] via 172.16.34.218, 00:55:17, Serial0/0/0
        [90/2682112] via 172.16.34.214, 00:55:17, Serial0/0/1
C       172.16.34.64/26 is directly connected, GigabitEthernet0/1
L       172.16.34.126/32 is directly connected, GigabitEthernet0/1
D       172.16.34.128/27 [90/2170112] via 172.16.34.218, 00:55:19, Serial0/0/0
D       172.16.34.160/27 [90/2170112] via 172.16.34.214, 00:55:17, Serial0/0/1
C       172.16.34.192/29 is directly connected, GigabitEthernet0/0
L       172.16.34.198/32 is directly connected, GigabitEthernet0/0
D       172.16.34.200/29 [90/2170112] via 172.16.34.214, 00:55:17, Serial0/0/1
S       172.16.34.208/30 is directly connected, Serial0/0/1
C       172.16.34.212/30 is directly connected, Serial0/0/1
L       172.16.34.213/32 is directly connected, Serial0/0/1
C       172.16.34.216/30 is directly connected, Serial0/0/0
L       172.16.34.217/32 is directly connected, Serial0/0/0
D       172.16.34.220/30 [90/2681856] via 172.16.34.218, 00:55:19, Serial0/0/0
D       172.16.34.224/30 [90/2681856] via 172.16.34.214, 00:55:17, Serial0/0/1
D       172.16.34.228/30 [90/2681856] via 172.16.34.218, 00:55:18, Serial0/0/0
D       172.16.34.232/30 [90/2681856] via 172.16.34.214, 00:55:17, Serial0/0/1
```

Figure 4.5: GPL router routing table

```

SL_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 17 subnets, 5 masks
D    172.16.34.0/26 [90/2170112] via 172.16.34.222, 00:55:17, Serial0/0/1
D    172.16.34.64/26 [90/2170112] via 172.16.34.217, 00:55:19, Serial0/0/0
C    172.16.34.128/27 is directly connected, GigabitEthernet0/0
L    172.16.34.158/32 is directly connected, GigabitEthernet0/0
D    172.16.34.160/27 [90/2682112] via 172.16.34.229, 00:55:17, Serial0/1/0
    [90/2682112] via 172.16.34.222, 00:55:17, Serial0/0/1
    [90/2682112] via 172.16.34.217, 00:55:17, Serial0/0/0
D    172.16.34.192/29 [90/2170112] via 172.16.34.217, 00:55:19, Serial0/0/0
D    172.16.34.200/29 [90/2682112] via 172.16.34.229, 00:55:17, Serial0/1/0
    [90/2682112] via 172.16.34.222, 00:55:17, Serial0/0/1
    [90/2682112] via 172.16.34.217, 00:55:17, Serial0/0/0
S    172.16.34.208/30 is directly connected, Serial0/1/0
D    172.16.34.212/30 [90/2681856] via 172.16.34.217, 00:55:19, Serial0/0/0
C    172.16.34.216/30 is directly connected, Serial0/0/0
L    172.16.34.218/32 is directly connected, Serial0/0/0
C    172.16.34.220/30 is directly connected, Serial0/0/1
L    172.16.34.221/32 is directly connected, Serial0/0/1
D    172.16.34.224/30 [90/2681856] via 172.16.34.222, 00:55:17, Serial0/0/1
C    172.16.34.228/30 is directly connected, Serial0/1/0
L    172.16.34.230/32 is directly connected, Serial0/1/0
D    172.16.34.232/30 [90/2681856] via 172.16.34.229, 00:55:18, Serial0/1/0

```

Figure 4.6: SL router routing table

```

NL_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 16 subnets, 5 masks
C    172.16.34.0/26 is directly connected, GigabitEthernet0/0
L    172.16.34.62/32 is directly connected, GigabitEthernet0/0
D    172.16.34.64/26 [90/2682112] via 172.16.34.221, 00:55:17, Serial0/0/1
    [90/2682112] via 172.16.34.226, 00:55:17, Serial0/0/0
D    172.16.34.128/27 [90/2170112] via 172.16.34.221, 00:55:17, Serial0/0/1
D    172.16.34.160/27 [90/2170112] via 172.16.34.226, 00:55:19, Serial0/0/0
D    172.16.34.192/29 [90/2682112] via 172.16.34.221, 00:55:17, Serial0/0/1
    [90/2682112] via 172.16.34.226, 00:55:17, Serial0/0/0
D    172.16.34.200/29 [90/2170112] via 172.16.34.226, 00:55:19, Serial0/0/0
S    172.16.34.208/30 is directly connected, Serial0/0/1
D    172.16.34.212/30 [90/2681856] via 172.16.34.226, 00:55:17, Serial0/0/0
D    172.16.34.216/30 [90/2681856] via 172.16.34.221, 00:55:17, Serial0/0/1
C    172.16.34.220/30 is directly connected, Serial0/0/1
L    172.16.34.222/32 is directly connected, Serial0/0/1
C    172.16.34.224/30 is directly connected, Serial0/0/0
L    172.16.34.225/32 is directly connected, Serial0/0/0
D    172.16.34.228/30 [90/2681856] via 172.16.34.221, 00:55:17, Serial0/0/1
D    172.16.34.232/30 [90/2681856] via 172.16.34.226, 00:55:17, Serial0/0/0

```

Figure 4.7: NL router routing table

```

IOT_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

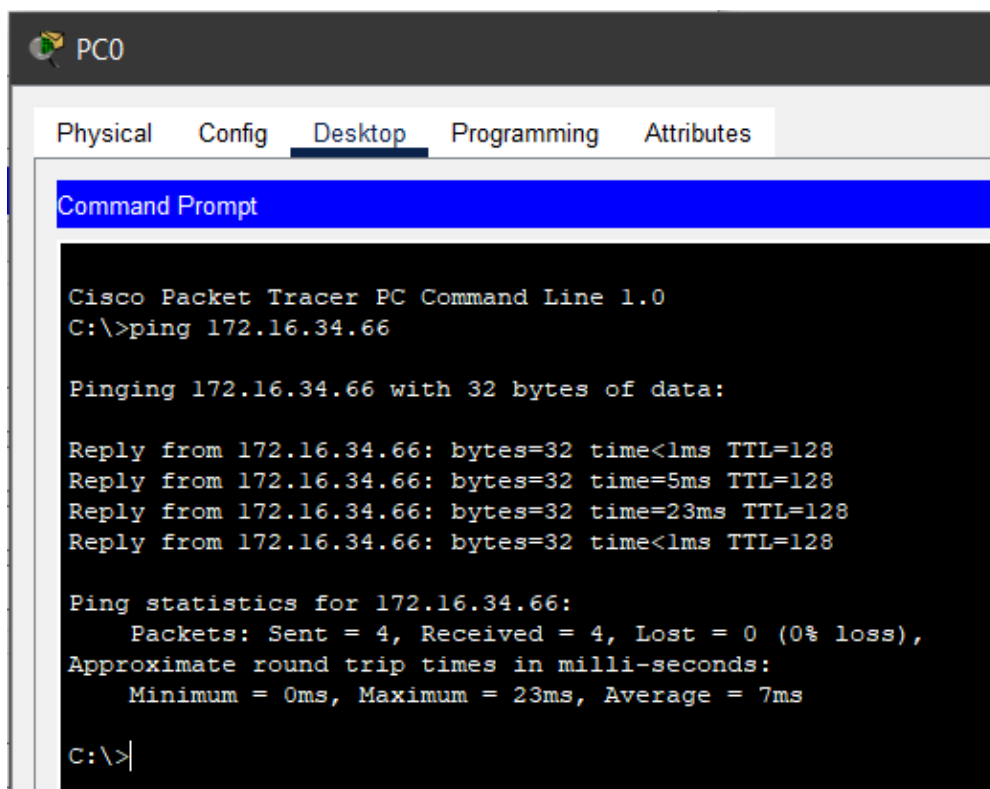
    172.16.0.0/16 is variably subnetted, 18 subnets, 5 masks
D       172.16.34.0/26 [90/2170112] via 172.16.34.225, 00:55:19, Serial0/0/0
D       172.16.34.64/26 [90/2170112] via 172.16.34.213, 00:55:17, Serial0/0/1
D       172.16.34.128/27 [90/2682112] via 172.16.34.233, 00:55:17, Serial0/1/1
           [90/2682112] via 172.16.34.225, 00:55:17, Serial0/0/0
           [90/2682112] via 172.16.34.213, 00:55:17, Serial0/0/1
C       172.16.34.160/27 is directly connected, GigabitEthernet0/1
L       172.16.34.190/32 is directly connected, GigabitEthernet0/1
D       172.16.34.192/29 [90/2170112] via 172.16.34.213, 00:55:17, Serial0/0/1
C       172.16.34.200/29 is directly connected, GigabitEthernet0/0
L       172.16.34.206/32 is directly connected, GigabitEthernet0/0
S       172.16.34.208/30 is directly connected, Serial0/1/1
C       172.16.34.212/30 is directly connected, Serial0/0/1
L       172.16.34.214/32 is directly connected, Serial0/0/1
D       172.16.34.216/30 [90/2681856] via 172.16.34.213, 00:55:17, Serial0/0/1
D       172.16.34.220/30 [90/2681856] via 172.16.34.225, 00:55:17, Serial0/0/0
C       172.16.34.224/30 is directly connected, Serial0/0/0
L       172.16.34.226/32 is directly connected, Serial0/0/0
D       172.16.34.228/30 [90/2681856] via 172.16.34.233, 00:55:17, Serial0/1/1
C       172.16.34.232/30 is directly connected, Serial0/1/1
L       172.16.34.234/32 is directly connected, Serial0/1/1

```

Figure 4.8: IOT router routing table

End-to-End Connection

Ping from	Ping to	Status
PC 0	PC 1	Successful
	PC 16	Successful
	PC 4	Successful
	PC 8	Successful
	PC 18	Successful
	PC 12	Successful
	Server	Successful



The screenshot shows the PC0 interface in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of the command 'ping 172.16.34.66'. The output indicates that the ping was successful, with four replies received from 172.16.34.66. The ping statistics show that all packets were received with 0% loss, and the average round trip time was 7ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.34.66

Pinging 172.16.34.66 with 32 bytes of data:

Reply from 172.16.34.66: bytes=32 time<1ms TTL=128
Reply from 172.16.34.66: bytes=32 time=5ms TTL=128
Reply from 172.16.34.66: bytes=32 time=23ms TTL=128
Reply from 172.16.34.66: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.34.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 23ms, Average = 7ms

C:\>|
```

Figure 4.9: PC0 to PC1 ping


```
C:\>ping 172.16.34.196

Pinging 172.16.34.196 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.196: bytes=32 time=41ms TTL=127
Reply from 172.16.34.196: bytes=32 time=3ms TTL=127
Reply from 172.16.34.196: bytes=32 time=31ms TTL=127

Ping statistics for 172.16.34.196:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 41ms, Average = 25ms

C:\>|
```

Figure 4.10: PC 0 to PC 16 ping

```
C:\>ping 172.16.34.192

Pinging 172.16.34.192 with 32 bytes of data:

Reply from 172.16.34.126: bytes=32 time<1ms TTL=255
Reply from 172.16.34.126: bytes=32 time<1ms TTL=255
Reply from 172.16.34.126: bytes=32 time<1ms TTL=255
Reply from 172.16.34.126: bytes=32 time=1ms TTL=255

Ping statistics for 172.16.34.192:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

Figure 4.11: PC 0 to PC 4 ping

```
C:\>ping 172.16.34.1

Pinging 172.16.34.1 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.1: bytes=32 time=2ms TTL=125
Reply from 172.16.34.1: bytes=32 time=34ms TTL=125
Reply from 172.16.34.1: bytes=32 time=34ms TTL=125

Ping statistics for 172.16.34.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 34ms, Average = 23ms
```

Figure 4.12: PC 0 to PC 8 ping

```
C:\>ping 172.16.34.203

Pinging 172.16.34.203 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.203: bytes=32 time=57ms TTL=126
Reply from 172.16.34.203: bytes=32 time=68ms TTL=126
Reply from 172.16.34.203: bytes=32 time=54ms TTL=126

Ping statistics for 172.16.34.203:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 54ms, Maximum = 68ms, Average = 59ms
```

Figure 4.13: PC 0 to PC 18 ping

```
C:\>ping 172.16.34.161

Pinging 172.16.34.161 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.161: bytes=32 time=1ms TTL=126
Reply from 172.16.34.161: bytes=32 time=1ms TTL=126
Reply from 172.16.34.161: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.34.161:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Figure 4.14: PC 0 to PC 12 ping

```
C:\>ping 172.16.34.209

Pinging 172.16.34.209 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.209: bytes=32 time=3ms TTL=125
Reply from 172.16.34.209: bytes=32 time=85ms TTL=125
Reply from 172.16.34.209: bytes=32 time=60ms TTL=125

Ping statistics for 172.16.34.209:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 85ms, Average = 49ms
```

Figure 4.15: PC 0 to Server ping

Reflection

This task provided practical experience with routing protocols and their configuration. We learned the differences between static and dynamic routing and their respective use cases. The task also emphasized the importance of routing in ensuring network connectivity and the ability to troubleshoot routing issues.

Task 5

Topology

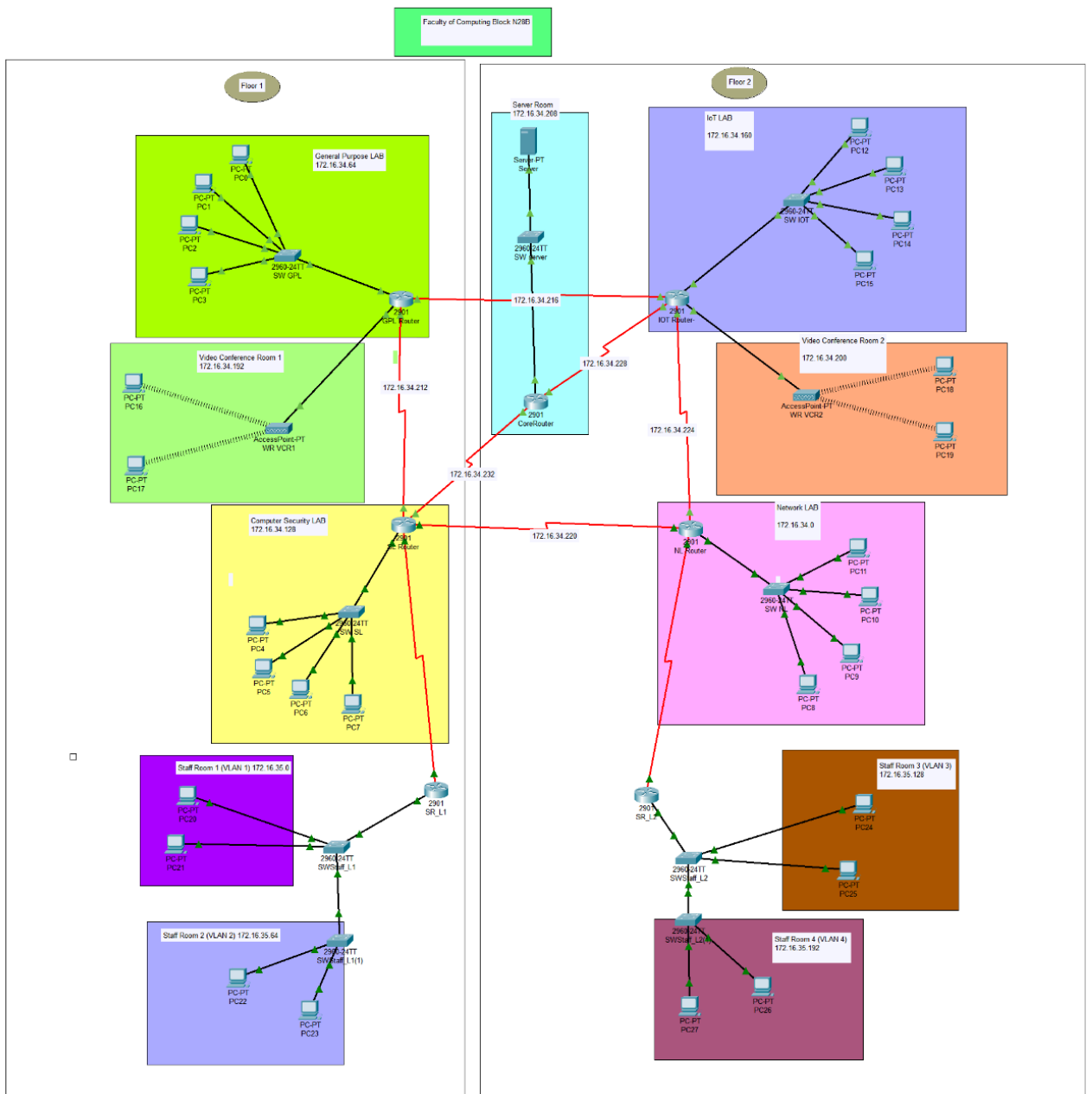


Figure 5.1: Faculty building topology in packet tracer

Subnetting

Area	Number of Hosts
Staff room 1 (VLAN1)	50
Staff room 2 (VLAN 2)	50
Staff room 3 (VLAN 3)	50
Staff room 4 (VLAN 4)	50
WAN SR_L1_SL	2
WAN SR_L2_NL	2

Calculation operation

172	16	35	0
10101100	.00010000	.00100001	1.00000000
	NP		HP
VLAN1=VLAN2=VLAN3=VLAN4=50			
$2^x = 50$			
$x = 6$			
WAN SR-L1-SL = WAN SR-L2-NL			
$2^x = 2$			
$x = 2$			
10101100.00010000.00100001.00000000 [0]			
$S_0 = \text{VLAN1}$	.00	111111	[63]
	.01	000000	[64]
$S_1 = \text{VLAN2}$	.01	111111	[127]
	.10	000000	[128]
$S_2 = \text{VLAN3}$	.10	111111	[191]
	.11	000000	[192]
$S_3 = \text{VLAN4}$	.11	111111	[255]
10101100.00001000.00100010.11101100 [236]			
$S_4 = \text{WAN SR-L1-SL}$	.111011	1111	[239]
	.111100	0000	[240]
$S_5 = \text{WAN SR-L2-SL}$	.111100	1111	[243]
	.111101	0000	[244]
$S_6 = \text{WAN CLOUD}$	.111101	1111	[247]
	.111110	0000	[248]

Figure 5.2: Subnetting Calculation

Details Range IP Address

Subnet	Area	Network Address	Broadcast Address	Usable Range IP Address
0	Network lab	172.16.34.0	172.16.34.63	172.16.34.1-172.16.34.62
1	General Purpose lab	172.16.34.64	172.16.34.127	172.16.34.65-172.16.34.126
2	Computer Security lab	172.16.34.128	172.16.34.159	172.16.34.129-172.16.34.158
3	IoT lab	172.16.34.160	172.16.34.191	172.16.34.161-172.16.34.190
4	Video conferencing rooms 1	172.16.34.192	172.16.34.199	172.16.34.193-172.16.34.198
5	Video conferencing rooms 2	172.16.34.200	172.16.34.207	172.16.34.201-172.16.34.206
6	Server room	172.16.34.208	172.16.34.211	172.16.34.209 - 172.16.34.210
7	WAN GPL - SL	172.16.34.212	172.16.34.215	172.16.34.213 - 172.16.34.214
8	WAN GPL - IOT	172.16.34.216	172.16.34.219	172.16.34.217 - 172.16.34.218
9	WAN SL - NL	172.16.34.220	172.16.34.223	172.16.34.221 - 172.16.34.222
10	WAN NL - IOT	172.16.34.224	172.16.34.227	172.16.34.225 - 172.16.34.226
11	WAN SR - IOT	172.16.34.228	172.16.34.231	172.16.34.229 - 172.16.34.230
12	WAN SR - SL	172.16.34.232	172.16.34.235	172.16.34.233 - 172.16.34.234
13	Staff Room 1	172.16.35.0	172.16.35.63	172.16.35.1 - 172.16.35.62
14	Staff Room 2	172.16.35.64	172.16.35.127	172.16.35.65 - 172.16.35.126

15	Staff Room 3	172.16.35.128	172.16.35.191	172.16.35.129 - 172.16.35.190
16	Staff Room 4	172.16.35.192	172.16.35.255	172.16.35.193 - 172.16.35.254
17	WAN SR_L1_SL	172.16.34.236	172.16.34.239	172.16.34.237 - 172.16.34.238
18	WAN SR_L2_SL	172.16.34.240	172.16.34.243	172.16.34.241 - 172.16.34.242
19	Cloud	172.16.34.244	172.16.34.247	172.16.34.245 - 172.16.34.246

IP Address assignment

- Staff room 1

Device	Interface	IP Address	Gateway Address	Subnet mask
PC20	fa0	172.16.35.1	172.16.35.62	255.255.255.192
PC21	fa0	172.16.35.2	172.16.35.62	255.255.255.192

- Staff room 2

Device	Interface	IP Address	Gateway Address	Subnet mask
PC20	fa0	172.16.35.65	172.16.35.126	255.255.255.192
PC21	fa0	172.16.35.66	172.16.35.126	255.255.255.192

- Staff room 3

Device	Interface	IP Address	Gateway Address	Subnet mask
PC20	fa0	172.16.35.129	172.16.35.190	255.255.255.192
PC21	fa0	172.16.35.130	172.16.35.190	255.255.255.192

- Staff room 4

Device	Interface	IP Address	Gateway Address	Subnet mask
PC20	fa0	172.16.35.193	172.16.35.254	255.255.255.192
PC21	fa0	172.16.35.194	172.16.35.254	255.255.255.192

Configuration

On Switch SWStaff_L1

Unset

```
interface fa0/1
switchport mode access
switchport access vlan 10
interface fa0/2
switchport mode access
switchport access vlan 10
exit
interface fa0/3
switchport mode trunk
interface g0/1
switchport mode trunk
exit
```

On Switch SWStaff_L1

Unset

```
interface fa0/1
switchport mode access
switchport access vlan 20
interface fa0/2
switchport mode access
switchport access vlan 20
exit
interface fa0/3
switchport mode trunk
exit
```

On Switch SWStaff_L2

Unset

```
interface fa0/4
switchport mode access
switchport access vlan 30
interface fa0/3
switchport mode access
switchport access vlan 30
exit
interface fa0/1
switchport mode trunk
interface g0/1
switchport mode trunk
exit
```

On Core router

Unset

```
interface serial 0/0/0
ip address 172.16.34.246 255.255.255.252
no shutdown
exit
ip route 0.0.0.0 0.0.0.0 serial 0/0/0
router eigrp 10
redistribute static
exit
```

On Switch SWStaff_L2

Unset

```
interface fa0/1
switchport mode access
switchport access vlan 40
interface fa0/2
switchport mode access
switchport access vlan 40
exit
interface fa0/3
switchport mode trunk
exit
```

On Cloud

Unset

```
hostname Cloud
interface se0/0/0
ip address 172.16.34.245 255.255.255.252
no shutdown
exit
ip route 172.16.0.0 255.255.0.0 Serial0/0/0
```

On Router SR_L1

Unset

```
enable
configure terminal
interface g0/0.10
encapsulation dot1Q 10
ip address 172.16.35.62 255.255.255.192
exit
interface g0/0.20
encapsulation dot1Q 20
ip address 172.16.35.126 255.255.255.192
exit
ip route 0.0.0.0 0.0.0.0 serial 0/0/0
```

On Router SL

Unset

```
enable
configuration terminal
ip route 172.16.35.0 255.255.255.192 serial 0/0/0
ip route 172.16.35.64 255.255.255.192 serial 0/0/0
router eigrp 10
redistribute static
```

On Router SR_L2

Unset

```
enable
configure terminal
interface g0/0.30
encapsulation dot1Q 30
ip address 172.16.35.190 255.255.255.192
exit
enable
configure terminal
interface g0/0.40
encapsulation dot1Q 40
ip address 172.16.35.254 255.255.255.192
exit
ip route 0.0.0.0 0.0.0.0 serial 0/0/1
```

On Router NL

Unset

```
enable
configuration terminal
ip route 172.16.34.192 255.255.255.192 serial 0/1/0
ip route 172.16.35.128 255.255.255.192 serial 0/1/0
router eigrp 10
redistribute static
```

VLAN Table

```
SWStaff_L1>enable
SWStaff_L1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	vlan1	active	Fa0/1, Fa0/2
20	vlan2	active	
99	management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Figure 5.3: Staff Room 1 (VLAN 1) table

```
SWStaff_L2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	vlan10	active	
20	vlan20	active	Fa0/1, Fa0/2
99	management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Figure 5.4: Staff Room 2 (VLAN 2) table

```
Switch>enable
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
30	vlan30	active	Fa0/3, Fa0/4
40	vlan40	active	
99	management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Figure 5.5: Staff Room 3 (VLAN 3) table


```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
30	vlan30	active	
40	vlan40	active	Fa0/1, Fa0/2
99	management	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Figure 5.6: Staff Room 4 (VLAN 4) table

End-to-End Connection

Ping from	Ping to	Status
PC 0	PC 1	Successful
	PC 16	Successful
	PC 4	Successful
	PC 8	Successful
	PC 18	Successful
	PC 12	Successful
	Server	Successful
	PC 20	Successful
	PC 22	Successful
	PC 24	Successful
	PC 26	Successful

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.34.66

Pinging 172.16.34.66 with 32 bytes of data:

Reply from 172.16.34.66: bytes=32 time<1ms TTL=128
Reply from 172.16.34.66: bytes=32 time<1ms TTL=128
Reply from 172.16.34.66: bytes=32 time<1ms TTL=128
Reply from 172.16.34.66: bytes=32 time=10ms TTL=128

Ping statistics for 172.16.34.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Figure 5.7: Ping to PC1 test

```

C:\>ping 172.16.34.195

Pinging 172.16.34.195 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.195: bytes=32 time=16ms TTL=127
Reply from 172.16.34.195: bytes=32 time=1ms TTL=127
Reply from 172.16.34.195: bytes=32 time=25ms TTL=127

Ping statistics for 172.16.34.195:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 25ms, Average = 14ms

```

Figure 5.8: Ping to PC16 test

```

C:\>ping 172.16.34.129

Pinging 172.16.34.129 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.129: bytes=32 time=1ms TTL=126
Reply from 172.16.34.129: bytes=32 time=1ms TTL=126
Reply from 172.16.34.129: bytes=32 time=2ms TTL=126

Ping statistics for 172.16.34.129:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

```

Figure 5.9: Ping to PC4

```

C:\>ping 172.16.34.1

Pinging 172.16.34.1 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.1: bytes=32 time=2ms TTL=125
Reply from 172.16.34.1: bytes=32 time=2ms TTL=125
Reply from 172.16.34.1: bytes=32 time=31ms TTL=125

Ping statistics for 172.16.34.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 31ms, Average = 11ms

```

Figure 5.10: Ping to PC8

```

C:\>ping 172.16.34.201

Pinging 172.16.34.201 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.201: bytes=32 time=30ms TTL=126
Reply from 172.16.34.201: bytes=32 time=4ms TTL=126
Reply from 172.16.34.201: bytes=32 time=237ms TTL=126

Ping statistics for 172.16.34.201:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 237ms, Average = 90ms

```

Figure 5.11: Ping to PC18

```

C:\>ping 172.16.34.161

Pinging 172.16.34.161 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.161: bytes=32 time=24ms TTL=126
Reply from 172.16.34.161: bytes=32 time=1ms TTL=126
Reply from 172.16.34.161: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.34.161:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 24ms, Average = 8ms

```

Figure 5.12: Ping to PC12

```

C:\>ping 172.16.34.209

Pinging 172.16.34.209 with 32 bytes of data:

Request timed out.
Reply from 172.16.34.209: bytes=32 time=3ms TTL=125
Reply from 172.16.34.209: bytes=32 time=3ms TTL=125
Reply from 172.16.34.209: bytes=32 time=3ms TTL=125

Ping statistics for 172.16.34.209:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3ms, Average = 3ms

```

Figure 5.13: Ping to Server

```
C:\>ping 172.16.35.1

Pinging 172.16.35.1 with 32 bytes of data:

Request timed out.
Reply from 172.16.35.1: bytes=32 time=2ms TTL=125
Reply from 172.16.35.1: bytes=32 time=57ms TTL=125
Reply from 172.16.35.1: bytes=32 time=39ms TTL=125

Ping statistics for 172.16.35.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 57ms, Average = 32ms
```

Figure 5.14: Ping to PC20

```
C:\>ping 172.16.35.65

Pinging 172.16.35.65 with 32 bytes of data:

Request timed out.
Reply from 172.16.35.65: bytes=32 time=2ms TTL=125
Reply from 172.16.35.65: bytes=32 time=17ms TTL=125
Reply from 172.16.35.65: bytes=32 time=2ms TTL=125

Ping statistics for 172.16.35.65:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 17ms, Average = 7ms
```

Figure 5.15: Ping to PC22

```
C:\>ping 172.16.35.129

Pinging 172.16.35.129 with 32 bytes of data:

Request timed out.
Reply from 172.16.35.129: bytes=32 time=3ms TTL=124
Reply from 172.16.35.129: bytes=32 time=3ms TTL=124
Reply from 172.16.35.129: bytes=32 time=41ms TTL=124

Ping statistics for 172.16.35.129:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 41ms, Average = 15ms
```

Figure 5.16: Ping to PC24

```
C:\>ping 172.16.35.193

Pinging 172.16.35.193 with 32 bytes of data:

Request timed out.
Reply from 172.16.35.193: bytes=32 time=4ms TTL=124
Reply from 172.16.35.193: bytes=32 time=4ms TTL=124
Reply from 172.16.35.193: bytes=32 time=3ms TTL=124

Ping statistics for 172.16.35.193:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 4ms, Average = 3ms
```

Figure 5.17: Ping to PC6

Reflection

This task introduced us to the concept of VLANs and their implementation in a network. We learned how VLANs can be used to segment network traffic and improve security and performance. The task also highlighted the importance of proper VLAN configuration and management in a complex network environment.

Task 6

DHCP Configuration

- General Purpose Lab:

The screenshot shows the Cisco Server configuration window with the 'Services' tab selected. The 'DHCP' service is highlighted in the left sidebar. The main configuration area is titled 'DHCP' and shows settings for the 'GigabitEthernet1' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'GeneralPuporseLab'. The 'Default Gateway' is '172.16.34.126'. The 'DNS Server' is '8.8.8.8'. The 'Start IP Address' is '172.16.34.65'. The 'Subnet Mask' is '255.255.255.192'. The 'Maximum Number of Users' is '61'. The 'TFTP Server' is '0.0.0.0'. The 'WLC Address' is '0.0.0.0'. Below the configuration fields are 'Add', 'Save', and 'Remove' buttons. At the bottom, there is a table showing the configuration for the 'serverPool'.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	172.16.34.65	255.255.255.192	61	0.0.0.0	0.0.0.0

Figure 6.1: General Purpose Lab DHCP configuration

Video Conference Room 1:

The screenshot shows the 'Server' configuration window with the 'Services' tab selected. The 'DHCP' service is configured for the 'GigabitEthernet1' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'VCR1'. The 'Default Gateway' is '172.16.34.198'. The 'DNS Server' is '8.8.8.8'. The 'Start IP Address' is '172.16.34.193'. The 'Subnet Mask' is '255.255.255.248'. The 'Maximum Number of Users' is '5'. The 'TFTP Server' is '0.0.0.0'. The 'WLC Address' is '0.0.0.0'. Below the configuration fields, there are 'Add', 'Save', and 'Remove' buttons. At the bottom, there is a table showing the DHCP configuration for the 'GeneralPuporseLab' and 'serverPool'.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
GeneralPuporseLab	172.16.34.198	8.8.8.8	172.16.34.193	255.255.255.248	61	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.34.193	255.255.255.248	512	0.0.0.0	0.0.0.0

Figure 6.2: Video Conference Room 1 DHCP configuration

GPL router

```
Unset
enable
configure terminal
interface gigabitethernet 0/1
ip helper-address 172.16.34.209
interface gigabitethernet 0/0
ip helper-address 172.16.34.209
exit
```


IoT Lab:

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: GigabitEthernet1 Service: ☒ On ☐ Off

Pool Name: IOTLab

Default Gateway: 172.16.34.190

DNS Server: 8.8.8.8

Start IP Address: 172 16 34 161

Subnet Mask: 255 255 255 224

Maximum Number of Users: 29

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
VCR1	172.16.34....	8.8.8.8	172.16.34....	255.255.2...	5	0.0.0.0	0.0.0.0
GeneralPuporseLab	172.16.34....	8.8.8.8	172.16.34.65	255.255.2...	61	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.34....	255.255.2...	512	0.0.0.0	0.0.0.0

Figure 6.3: IoT Lab DHCP configuration

Video conference 2:

Server Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: GigabitEthernet1 Service: ☒ On ☐ Off

Pool Name: VRC2

Default Gateway: 172.16.34.206

DNS Server: 8.8.8.8

Start IP Address: 172 16 34 201

Subnet Mask: 255 255 255 248

Maximum Number of Users: 5

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server
IOTLab	172.16.34....	8.8.8.8	172.16.34.161	255.255.255.224	29	0.0.0.0

Figure 6.4: Video Conference Room 2 DHCP configuration

IOT_Router

```
Unset
enable
configure terminal
interface gigabitethernet 0/1
ip helper-address 172.16.34.209
interface gigabitethernet 0/0
ip helper-address 172.16.34.209
exit
```

Computer Security Lab:

The screenshot shows the 'Server' configuration window with the 'Services' tab selected. The 'DHCP' service is configured for the 'GigabitEthernet1' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'ComputerSecurityLab'. The 'Default Gateway' is '172.16.34.158'. The 'DNS Server' is '8.8.8.8'. The 'Start IP Address' is '172.16.34.129'. The 'Subnet Mask' is '255.255.255.224'. The 'Maximum Number of Users' is '29'. The 'TFTP Server' is '0.0.0.0'. The 'WLC Address' is '0.0.0.0'. Below the configuration fields are 'Add', 'Save', and 'Remove' buttons. A table lists the DHCP pools:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFT Serv
VCR2	172.16.34....	8.8.8.8	172.16.34.201	255.255.255.248	5	0.0.0.0
IOTLab	172.16.34....	8.8.8.8	172.16.34.161	255.255.255.224	29	0.0.0.0
VCR1	172.16.34....	8.8.8.8	172.16.34.193	255.255.255.248	5	0.0.0.0
GeneralPuporseLab	172.16.34....	8.8.8.8	172.16.34.65	255.255.255.192	61	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.34.208	255.255.255.252	512	0.0.0.0

Figure 6.5: Computer Security Lab DHCP configuration

SL_router

```
Unset
enable
configure terminal
interface gigabitethernet 0/0
ip helper-address 172.16.34.209
exit
```

Pool	Default	DNS	Start IP	Subnet	Max	TFTP	WLC
NetworkLab	172.16.34.62	8.8.8.8	172.16.34.1	255.255.255.192	61	0.0.0.0	0.0.0.0

Figure 6.6: Network Lab DHCP configuration

NL_router

```
Unset
enable
configure terminal
interface gigabitethernet 0/0
ip helper-address 172.16.34.209
exit
```

Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: GigabitEthernet1 Service: ☒ On ☐ Off

Pool Name: StaffRoom1

Default Gateway: 172.16.35.62

DNS Server: 8.8.8.8

Start IP Address: 172 16 35 1

Subnet Mask: 255 255 255 192

Maximum Number of Users: 61

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
StaffRoom1	172.16.35.62	8.8.8.8	172.16.35.1	255.255.255.192	61	0.0.0.0	0.0.0.0
NetworkLab	172.16.34.62	8.8.8.8	172.16.34.1	255.255.255.192	61	0.0.0.0	0.0.0.0

Figure 6.7: Staff Room 1 DHCP configuration

Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: GigabitEthernet1 Service: ☒ On ☐ Off

Pool Name: StaffRoom2

Default Gateway: 172.16.35.126

DNS Server: 8.8.8.8

Start IP Address: 172 16 35 65

Subnet Mask: 255 255 255 192

Maximum Number of Users: 61

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
StaffRoom2	172.16.35.126	8.8.8.8	172.16.35.65	255.255.255.192	61	0.0.0.0	0.0.0.0

Figure 6.8: Staff Room 2 DHCP configuration

Router SR_L1

```
Unset
enable
configure terminal
interface gigabitethernet 0/0.10
ip helper-address 172.16.34.209
exit
interface gigabitethernet 0/0.20
ip helper-address 172.16.34.209
exit
```

The screenshot shows the Cisco Configuration Assistant (CCA) interface with the 'Services' tab selected. On the left, a sidebar lists various services, with 'DHCP' highlighted. The main area is titled 'DHCP' and contains configuration fields for a DHCP pool named 'StaffRoom3'. The interface is set to 'GigabitEthernet1' and the service is turned 'On'. The configuration includes a default gateway of 172.16.35.190, a DNS server of 8.8.8.8, and a start IP address of 172.16.35.129 with a subnet mask of 255.255.255.192. The maximum number of users is set to 61. TFTP and WLC addresses are both set to 0.0.0.0. At the bottom, there are 'Add', 'Save', and 'Remove' buttons, and a table showing the configuration details.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
StaffRoom3	172.16.35.190	8.8.8.8	172.16.35.129	255.255.255.192	61	0.0.0.0	0.0.0.0

Figure 6.9: Staff Room 3 DHCP configuration

Server

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: GigabitEthernet1 Service: ☒ On ☐ Off

Pool Name: StaffRoom4

Default Gateway: 172.16.35.254

DNS Server: 8.8.8.8

Start IP Address : 172 16 35 193

Subnet Mask: 255 255 255 192

Maximum Number of Users : 61

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool	Default	DNS	Start in	Subnet	Max	TFTP	WLC

Figure 6.10 : Staff Room 4 DHCP configuration

Router SR_L2

```
Unset
enable
configure terminal
interface gigabitethernet g0/0.30
ip helper-address 172.16.34.209
exit
interface gigabitethernet 0/0.40
ip helper-address 172.16.34.209
exit
```

Security Configuration

All router

```
Unset
enable
configure terminal
no ip domain-lookup
enable secret class
line console 0
password cisco
login
line vty 0 4
password cisco
login
exit
service password-encryption
banner motd $ Authorized Users Only! $
exit
copy running-config startup-config
```

DHCP Message Exchange Process

In this example we will demonstrate how PC 14 as a DHCP client gets an IP address from a DHCP server and IoT router as relay agent to the DHCP server. Here we will explain each DORA process : Discover, Offer, Request and Acknowledge [1].

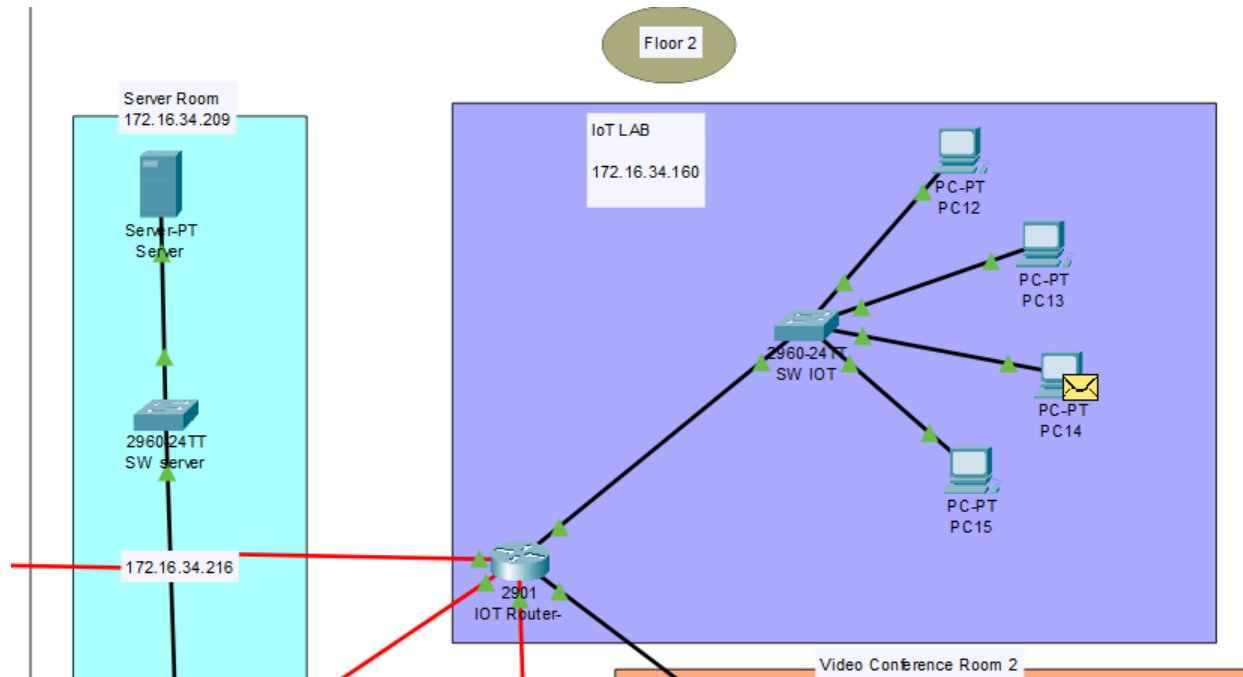


Figure 6.11: DHCP client send discover packet

- The DHCP client constructs a Discover packet and sends it out

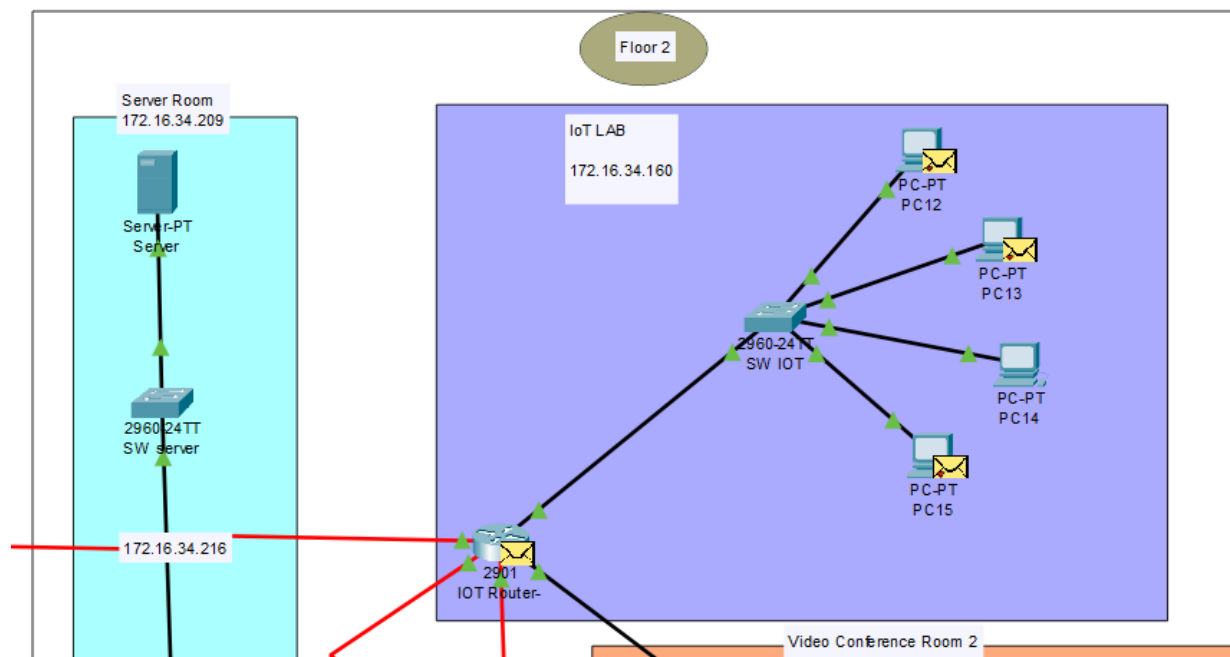


Figure 6.12: Relay agent send to helper address

- The packet is a DHCP packet. The DHCP server processes it.
- The relay agent received a DHCP Discover packet.

- The relay agent sends it to the helper's address since the packet matches the helper's criteria. The device forwards the packet to the helper address.

DHCP Offer

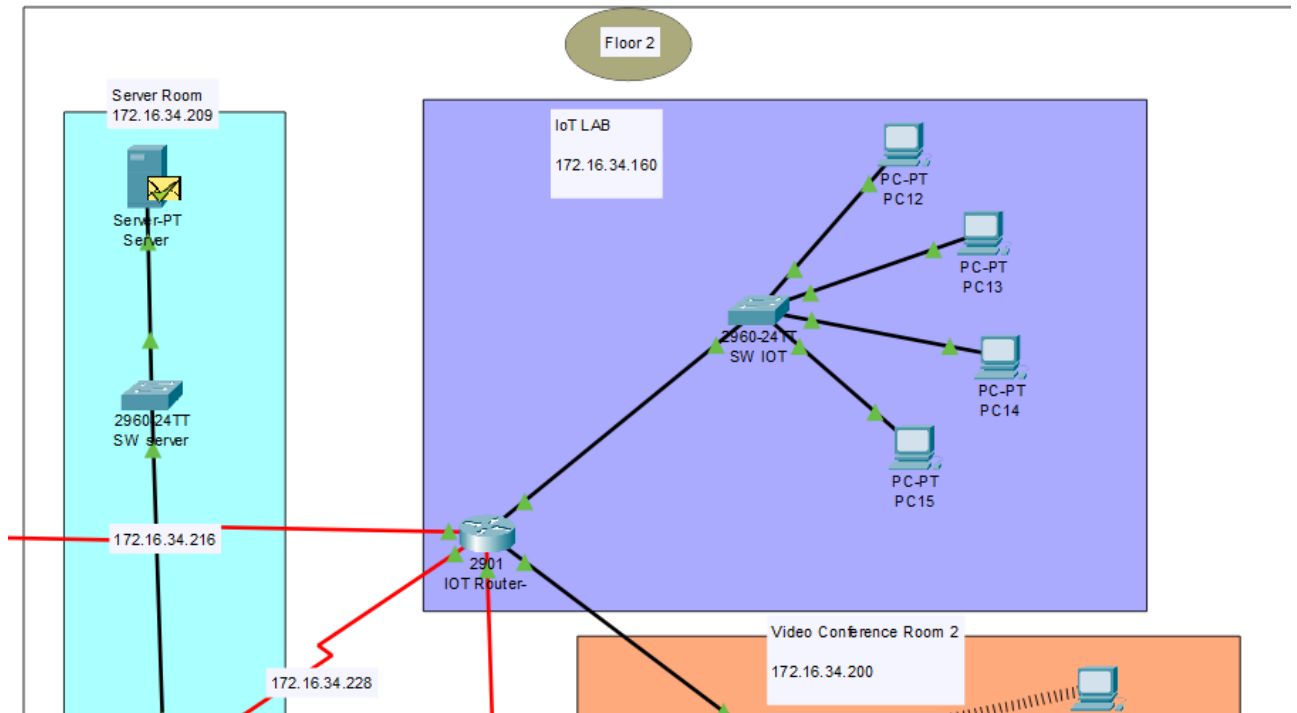


Figure 6.13: DHCP server sends offer packet

- The packet is a DHCP packet. The DHCP server processes it.
- The DHCP server received a DHCP Discover packet.
- The DHCP server does not have an existing binding to this host. It looks up DHCP pools for a new IP address.
- The DHCP server finds the following available IP address in the pool
- The DHCP sent out an offer packet to the client.

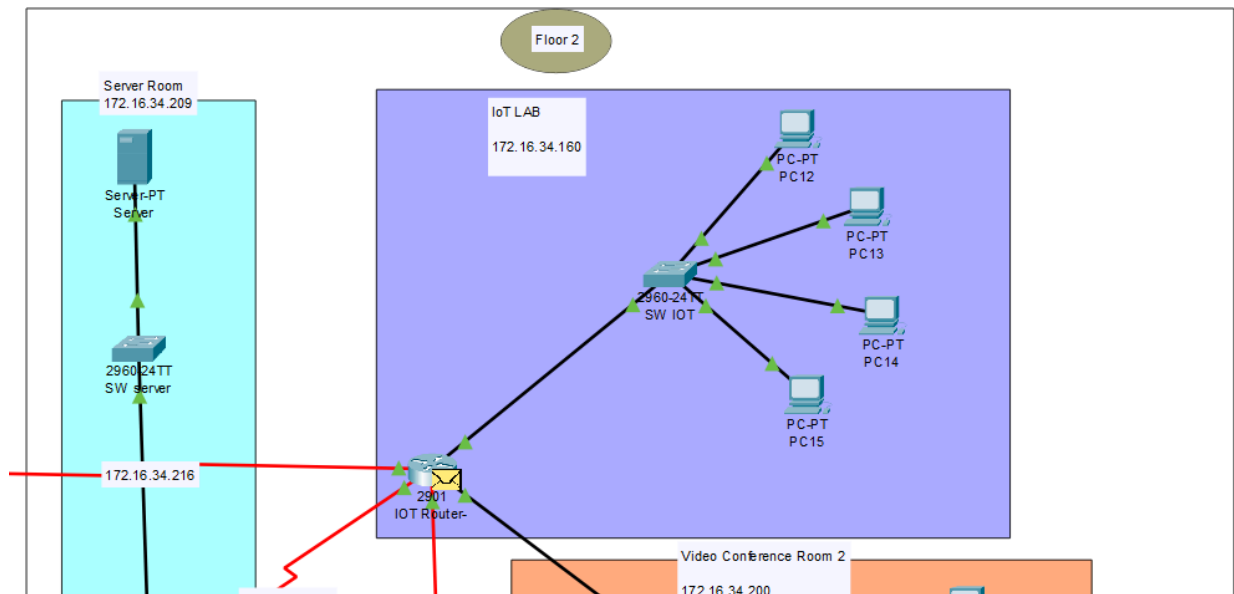


Figure 6.14: Relay agent forward reply to DHCP client

- The packet is a reply from the helper address. The device forwards the packet back to the original sender.

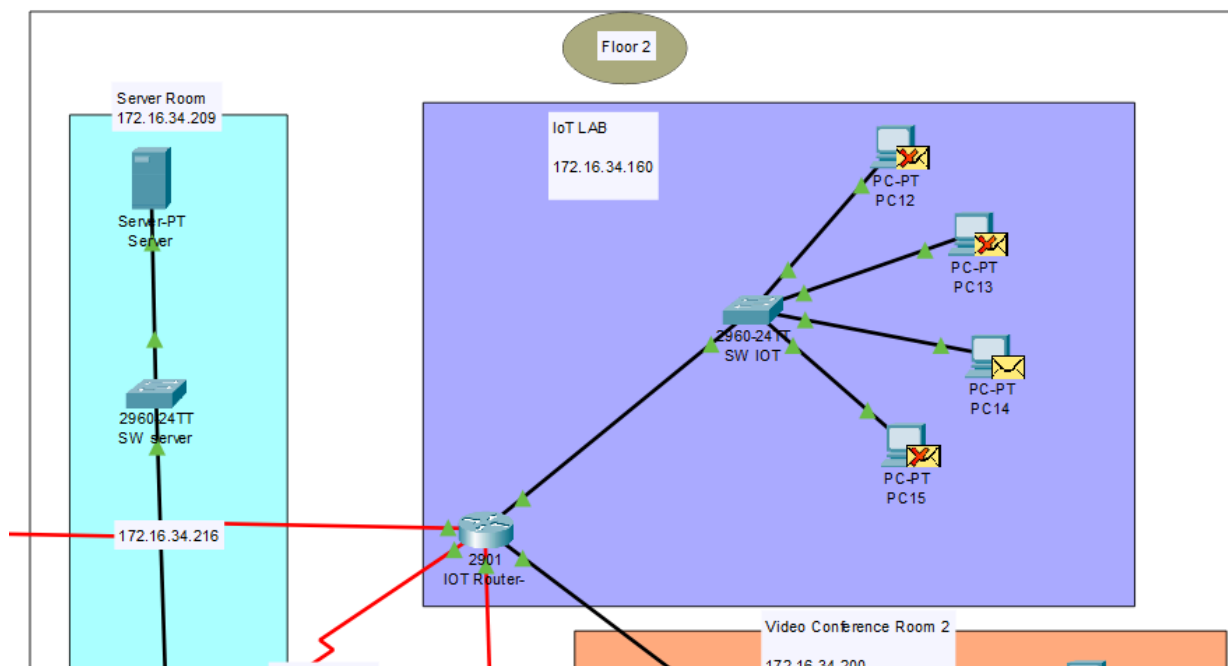


Figure 6.15: DHCP client sends out a request packet

- The packet is a DHCP packet. The DHCP client processes it.
- The DHCP client received a DHCP offer packet.
- The DHCP client sends out a DHCP request to the server.

DHCP Request

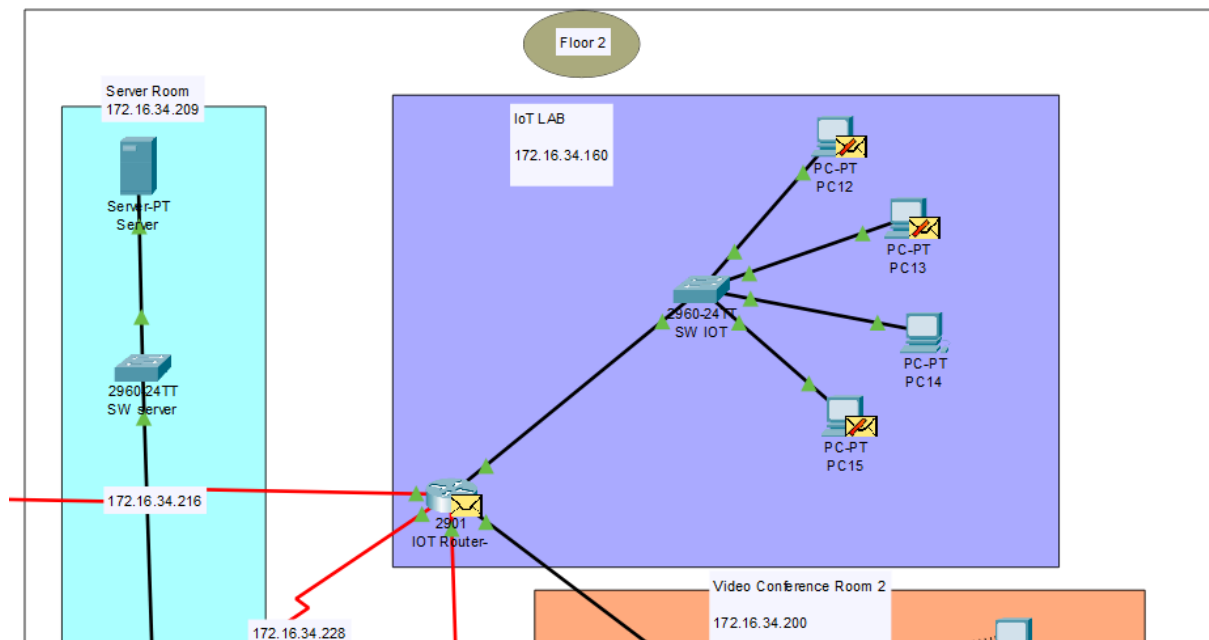


Figure 6.16: Relay agent forwards request packet to DHCP server

- The packet is a DHCP packet. The DHCP server processes it.
- The packet matches the helper criteria. The device forwards the packet to the helper address.
- The device looks up the destination IP address in the routing table.
- The routing table finds a routing entry to the destination IP address.
- The destination routing entry is a connected static route. The device sets the destination as the next hop.

DHCP Acknowledge

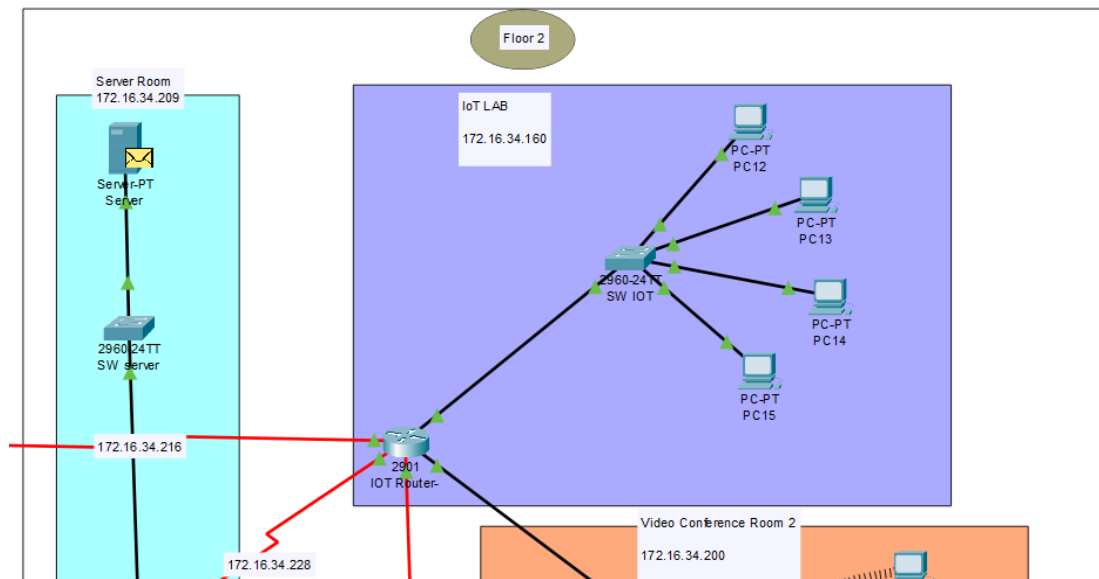


Figure 6.17: The DHCP server sends an acknowledgment packet

- The packet is a DHCP packet. The DHCP server processes it.
- The DHCP server received a DHCP Request packet.
- The DHCP server binds the requested IP address from the pool to the host's MAC address.
- The DHCP server sends out an Ack packet to the host.

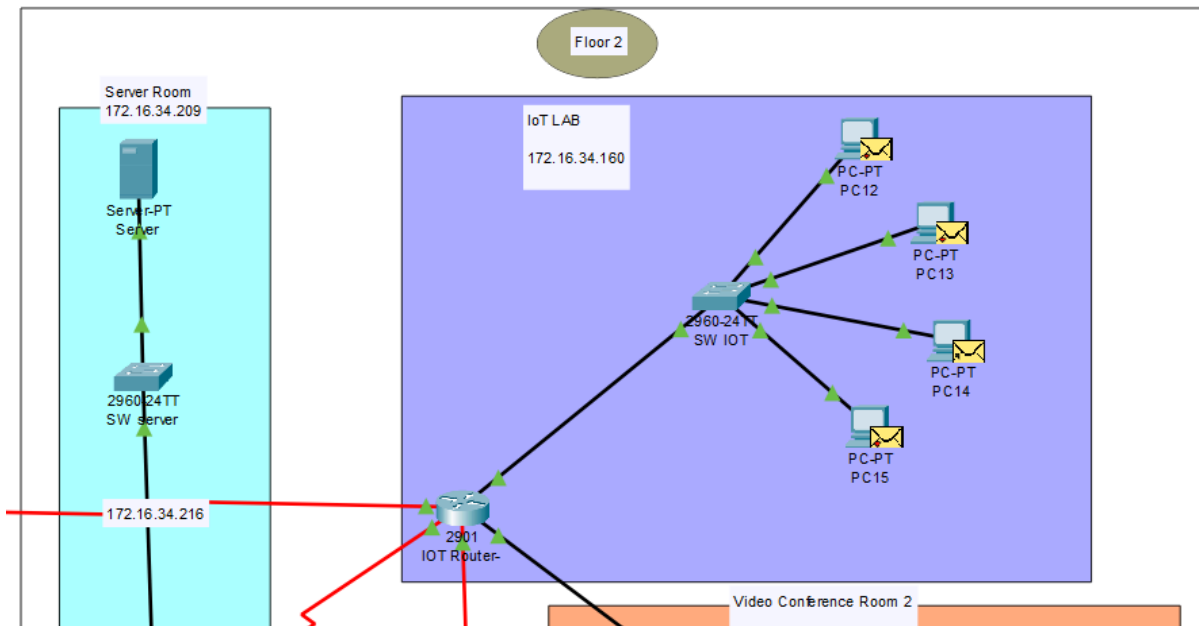


Figure 6.18: DHCP client receives acknowledgment packet

- The packet is a DHCP packet. The DHCP client processes it.
- The DHCP client received a DHCP acknowledge packet.
- The DHCP client receives an Ack packet and sets its IP address configuration.

The result

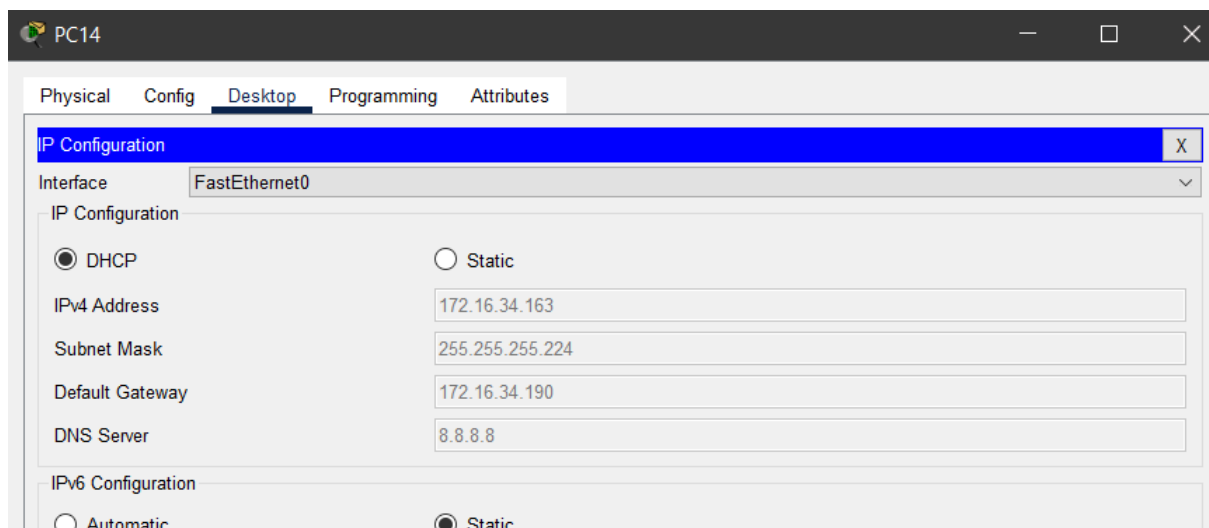


Figure 6.19: DHCP client sends IP address

End-to-End Connection

Ping from	Ping to	Status
PC 0	PC 1	Successful
	PC 16	Successful
	PC 4	Successful
	PC 8	Successful
	PC 18	Successful
	PC 12	Successful
	Server	Successful
	PC 20	Successful
	PC 22	Successful
	PC 24	Successful
	PC 26	Successful

PDU List Window										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC16	ICMP		0.000	N	1	(edit)	
	Successful	PC0	PC4	ICMP		0.000	N	2	(edit)	
	Successful	PC0	PC20	ICMP		0.000	N	3	(edit)	
	Successful	PC0	PC22	ICMP		0.000	N	4	(edit)	
	Successful	PC0	PC12	ICMP		0.000	N	5	(edit)	
	Successful	PC0	PC18	ICMP		0.000	N	6	(edit)	
	Successful	PC0	PC11	ICMP		0.000	N	7	(edit)	
	Successful	PC0	PC24	ICMP		0.000	N	8	(edit)	
	Successful	PC0	PC24	ICMP		0.000	N	9	(edit)	
	Successful	PC0	PC26	ICMP		0.000	N	10	(edit)	
	Successful	PC0	Server	ICMP		0.000	N	11	(edit)	
	Successful	PC0	Cloud	ICMP		0.000	N	12	(edit)	

Figure 6.20: PDU list

Reflection

We configured DHCP servers in our network to dynamically assign IP addresses to hosts in the labs. This ensured that all devices received IP addresses automatically, simplifying network management and reducing the risk of IP address conflicts.

Conclusion

We have successfully developed a detailed floor plan and network topology diagram for a two storey building, ensuring the functionality by using Cisco Packet Tracer. We gained a deeper understanding of network topology and the practical application of networking concepts. Next, we ensure there is no wasted IP address in our topology, this task enhances our understanding and subnetting. We learned the importance of proper IP address allocation and the potential issues that can arise from IP address conflict. We implement dynamic routing by using Enhanced Interior Gateway Routing Protocol (EIGRP) and check the end to end connection between PCs. This task provided practical experience with routing protocols and their configuration. We learned the differences between static and dynamic routing and their respective use cases.

Our strength is we already learn within the courses and have been given the chance to implement and do projects about network operation virtualization. By anticipating future needs and incorporating scalability, we ensured that our network can grow and adapt over time. But our weaknesses are the level of implementation and complexity of setup that required specialized knowledge and expertise to maintain and troubleshoot. We also need to make sure our groupmates have the necessary knowledge to utilize the network devices in the Cisco Packet Tracer.

Team Members and Responsibilities

Team Members	Task
Aidan Andrew Daniel	Task 1: Suggested group name and Describe Network lab and IOT lab (floor 2). Task 2: Lead by assessing the task and reviewing documentation Task 3: Set details range of IP address Task 4: Provide script for configuration for Floor 1 Task 5: Implementing VLAN and Routing Task 6: Introduction and conclusion Task 7: Team member Responsibility
Muhammad Amirul Azfar Bin Azhar	Task 1: Describe the security lab and general lab (floor 1). Task 2: Create topology by using a packet tracer Task 3: Set the IP address of each host in the topology Task 4: Configure the Floor 1 area using the script provided Task 5: Check connectivity and show the VLAN table Task 6: explain the DHCP process Task 7: Introduction and conclusion
Danial Nur Irfan bin Azeze	Task 1: Introduction and Conclusion Task 2: In charge of Introduction and Conclusion Task 3: Do the calculation for the subnetting Task 4: Configure the Floor 2 area using the script provided

	Task 5: Do Subnetting and document network assignment. Task 6: Check End to end connection and do Security configuration Task 7: Project background and formation the references and appendices
Muhammad Haziq Bin Saamsol	Task 1: Modify the Floor plan and describe the Video conference room. Task 2: Find and extract information about topology (implementation) Task 3: In charge of introduction and conclusion Task 4: Provide script for configuration for Floor 2 Task 5: Do Introduction and Conclusion Task 6: Configure DHCP Task 7: Combine all the tasks.

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Appendices

MEETING MINUTES 1 TASK 1 Group Network Ninjaz

DATE/TIME		20 March 2024 (11pm-12am)	
LOCATION		Telegram	
AGENDA		1. Group Name 2. Choose Floor plan 3. Task divide	
Meeting MC		Muhamad Haziq Bin Saamsol	
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Aidan Andrew Daniel	11.00	-	
Muhammad Amirul Azfar Bin Azhar	11.07	-	
Danial Nur Irfan bin Azeze	11.10	-	
Muhammad Haziq Bin Saamsol	11.03	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Group name	Aidan suggested the “NetworkNinjaz” for Group name.	Aidan
2	Choose Floor plan	Three different floor plans were shared from the previous semester's project and one was voted for this project's use.	All members
3	Floor plan modify	Using draw.io to modify floor plan according to the current case study	Haziq

MEETING MINUTES 2 TASK 1
Group Network Ninjaz

DATE/TIME		21 March 2024 (3pm-3.30am)	
LOCATION		Telegram	
AGENDA		1. Task division	
Meeting MC		Muhammad Amirul Azfar Bin Azhar	
ATTENDANCE			
NAME	TIME		REASON FOR ABSENCE
Aidan Andrew Daniel	3.00		-
Muhammad Amirul Azfar Bin Azhar	3.00		-
Danial Nur Irfan bin Azeze	3.00		-
Muhammad Haziq Bin Saamsol	3.00		-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Introduction and conclusion	Explain the overview of the floor plan and project	Danial
2	Describe floor 1	Describe Security lab and General Purpose lab	Azfar
3	Describe floor 2	Describe Network lab and IOT lab	Aidan
4	Describe video conference	Describe video conference	Haziq

MEETING MINUTES 1 TASK 2
Group Network Ninjaz

DATE/TIME		1 April 2024 (8PM - 10PM)	
LOCATION		Telegram	
AGENDA		1. Task division 2. Topology creation	
Meeting MC		Aidan Andrew Daniel	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Aidan Andrew Daniel		8.00	-
Muhammad Amirul Azfar Bin Azhar		8.30	-
Danial Nur Irfan bin Azeze		8.00	-
Muhammad Haziq Bin Saamsol		8.10	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Project plan	Haziq suggested how the topology should be	Haziq
2	Documentation: Prepare and lead	Aidan will incharge for leader in this task and take care in reviewing the progress	Aidan
3	Documentation: Introduction and conclusion	Daniel will incharge to do introduction and conclusion	Danial
4	Documentation: Implementation	Haziq will incharge to tell detailed about the topology	Haziq
5	Create topology	Azhar voluntarily to create the topology in packet tracer	Azhar
6	Topology creation idea	All members create their own topology and then compare with each other	All members

MEETING MINUTES 2 TASK 2
Group Network Ninjaz

DATE/TIME		3 April 2024 (10PM - 12AM)	
LOCATION		Telegram	
AGENDA		1. Topology creation 2. Google docs completion	
Meeting MC		Aidan Andrew Daniel	
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Aidan Andrew Daniel	10.00	-	
Muhammad Amirul Azfar Bin Azhar	10.00	-	
Danial Nur Irfan bin Azeze	10.00	-	
Muhammad Haziq Bin Saamsol	10.00	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Topology choosing	Decided to go with Haziq's topology.	All members
2	Topology review and modification	Review topology create in packet tracer and do some modification	Haziq and Azfar
3	Google docs completion	All member completed the google docs based on their own task chosen	All members

MEETING MINUTES 1 TASK 3
Group Network Ninjaz

DATE/TIME		15 April 2024 (8PM - 10PM)	
LOCATION		Telegram	
AGENDA		1. Task division	
Meeting MC		Danial Nur Irfan bin Azeze	
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Aidan Andrew Daniel	11.10 pm	-	
Muhammad Amirul Azfar Bin Azhar	11.10 pm	-	
Danial Nur Irfan bin Azeze	11.00 pm	-	
Muhammad Haziq Bin Saamsol	11.10 pm	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	List task	List all the task & prepare doc	Danial
2	Introduction and conclusion	Haziq will do introduction and conclusion	Haziq
3	Subnetting calculation	Daniel will do the working of subnetting	Daniel
4	Details Range IP address	Aidan will set the details Range IP address	Aidan
5	IP allocation	Azfar will allocate all host IP	Azfar

MEETING MINUTES 2 TASK 3
Group Network Ninjaz

DATE/TIME		18 April 2024 (10PM - 12AM)	
LOCATION		Telegram	
AGENDA		1. Review working subnetting	
Meeting MC		Danial Nur Irfan bin Azeze	
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Aidan Andrew Daniel	11.00 am	-	
Muhammad Amirul Azfar Bin Azhar	11.00 am	-	
Danial Nur Irfan bin Azeze	11.00 am	-	
Muhammad Haziq Bin Saamsol	11.00 am	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Subnetting validation	All member try do subnetting and compare the result	All members

MEETING MINUTES 1 TASK 4
Group Network Ninjaz

DATE/TIME		15 May 2024 (8PM - 10PM)	
LOCATION		Aidan’s room	
AGENDA		1. Making script for packet tracer 2. Configuring all the routing protocols	
Meeting MC		Aidan Andrew Daniel	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Aidan Andrew Daniel		8.00 pm	-
Muhammad Amirul Azfar Bin Azhar		8.30 pm	-
Danial Nur Irfan bin Azeze		8.30 pm	-
Muhammad Haziq Bin Saamsol		8.10 pm	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Floor configuration division	Haziq divided the floor for configuration purpose	Haziq
2	Deciding when to use static and dynamic routing	Aidan suggested where to implement static and dynamic routing	Aidan
3	Making script for configuration	Aidan & Haziq made the script to configure the routing protocols	Aidan & Haziq
4	Configuring the routing protocols	Danial & Azfar configured the routing protocols for both floor	Danial & Azfar
5	Verifying connection	Danial & Azfar verify the connection by pinging all devices	Danial & Azfar

MEETING MINUTES 1 TASK 5
Group Network Ninjaz

DATE/TIME		26 May 2024	
LOCATION		Telegram	
AGENDA		1. Subnetting host of staff room 2. Type Vlans 3. Assign task	
Meeting MC		Aidan Andrew Daniel	
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Aidan Andrew Daniel	8.00 pm	-	
Muhammad Amirul Azfar Bin Azhar	8.30 pm	-	
Danial Nur Irfan bin Azeze	8.30 pm	-	
Muhammad Haziq Bin Saamsol	8.10 pm	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Decide how much host in Staff room	Aidan give idea to set 50 usable host for each Staff room	Haziq
2	Decide what VLAN want to implement	Azhar told us try to do router on stick which is better than use inter VLAN which outdated	Azhar
3	VLAN and routing implementation	Aidan will do the VLAN configuration and the routing.	Aidan
4	Subnetting	Danial will do the calculation of the subnetting and document it	Danial
5	Verifying connection	Azhar will do the end-to-end verification after done the configuration	Azhar
6	Introduction and Conclusion	Haziq will do the Introduction and conclusion	Haziq

MEETING MINUTES 1 TASK 6
Group Network Ninjaz

DATE/TIME		14 June 2024	
LOCATION		Azhar Room	
AGENDA		1. Assign task 2. Router or server as DHCP server	
Meeting MC		Aidan Andrew Daniel	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Aidan Andrew Daniel		10.00 pm	-
Muhammad Amirul Azfar Bin Azhar		10:00 pm	-
Danial Nur Irfan bin Azeze		10.10 pm	-
Muhammad Haziq Bin Saamsol		10.02 pm	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Decide router or server as DHCP Server	Haziq suggest to use server as DHCP server since it more easy to configure	Haziq
2	Introduction and conclusion task	Aidan will do the Introduction and Conclusion task	Aidan
3	DHCP configuration	Haziq will do the DHCP configuration	Haziq
4	DHCP process	Azhar will do explanation how DHCP working	Azhar
5	Verifying connection and security configuration	Daniel will check connectivity and add security measure	Danial

MEETING MINUTES 1 TASK 7
Group Network Ninjaz

DATE/TIME		25 June 2024	
LOCATION		Aidan Room	
AGENDA		1. Assign task 2. Suggesting reflection	
Meeting MC		Haziq	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Aidan Andrew Daniel		8.00 pm	-
Muhammad Amirul Azfar Bin Azhar		8:10 pm	-
Danial Nur Irfan bin Azeze		8.10 pm	-
Muhammad Haziq Bin Saamsol		8.00 pm	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS & SUGGESTIONS	PERSON IN CHARGE & DATE
1	Task division	Haziq divided the task to complete the report	Haziq
2	Reflection for each task	Everyone suggested their reflection for each task	Everyone
3	Conclusion	Azfar will do the conclusion for the project	Azfar
4	Team members responsibilities	Aidan will analyse each members responsibilities to include in the task	Aidan
5	Combining all references and appendices	Danial will combine all references and appendices from previous task	Danial

Rubric Task 1

TASK 1	
ITEM	MARKS
Group name and members	2
Chosen floor plan.	
Clear description of floor plan chosen	4
Scale: with scale	0.5
Clear and readable. Has appropriate labels	1
Good and clear meeting minutes	1
Follow good report writing practice	1
TOTAL	9.5

Rubric Task 2

TASK 2	
ITEM	MARKS
Packet Tracer is done well [clear and readable]	3
Matches floor plan suggested in Task 1	2
Device and area clearly labelled	3
Good and clear meeting minutes	1
Follow good report writing practice	1
TOTAL	10

Rubric Task 3

TASK 3	
ITEM	MARKS
IP Addressing	
Use correct network address for group	1
Workings is provided clearly and labelled	4
IP division is appropriate and logical	1
All devices and host has been assigned a correct IP address and gateway address (as required)	2
Good and clear meeting minutes	1
Follow good report writing practice	1
TOTAL	10

Rubric Task 4

TASK 4	
ITEM	MARKS
Implementing Routing	
Clear markings on topology of routing protocol used	2
Clear configurations shown in report	3
Content of routing table from ALL routers clearly shown in report	2
End-to-end connection is achieved.*shown in report	1
End-to-end connection is achieved.*can be tested on pkt file	1
Follow good report writing practice	1
TOTAL	10

Rubric Task 5

TASK 5	
ITEM	MARKS
Implementing VLAN	
Staff rooms added as a requirement. (shown in report and PT)	2
IP assignments to staff rooms clearly shown	2
Clear VLAN configurations shown in report	2
Content of VLAN table clearly shown in report	1
End-to-end connection is achieved.*proof shown in report	2
Good and clear meeting minutes	1
TOTAL	10

Rubric Task 6

TASK 6	
ITEM	MARKS
Implementing DHCP	
Clear DHCP configurations shown in report	5
Clear examples of DHCP working (screenshot to report)	4
Good and clear meeting minutes	1
TOTAL	10

Rubric Task 7A

TASK 7A	
ITEM	MARKS
7A: GROUP REPORT	
Title page follows requirement	0.25
TOC clearly and correctly done	0.25
List of Figures - available, appropriate and correctly done	0.25
Introduction: done well, help with understanding, did not copy and paste	1
Project background clearly and correctly done	1
A compiled solution (all task) with reflections	6
Conclusion clearly and correctly done	1
References clearly and correctly done	1
Correctly formatted	0.25
Team meetings	
Team Members and responsibilities	1
Team meeting minutes	1
Appendices: complete with all the requirements as in Project document	1
Follow good report writing practice	1
TOTAL	15

Group discussion

